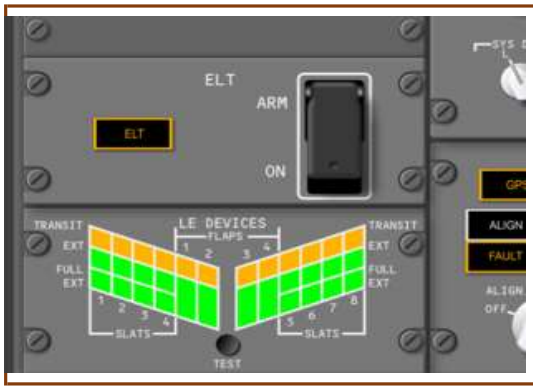


NNC: ELT

1

1

NNC: GPS

SHORT: ABORT RNP APPROACH
IAN may be continued
(tenzij: unable reqd nav perf-rnp)

nu ook ADS-B OUT fail (NNC useless)

if pos. shifts due jamming, evt oplossing:
NAV DATA -> GPS OFF, DME ON
nu gebruik je IRS (nog betrouwbaar)

1

2

NNC: ILS

do not fly ILS approach
both receivers are faulty
or
both IRS detected a fault

1

3

NNC: GLS

do not fly GLS approach
(GNSS landing system)

1

4

NNC: IRS ON DC

both IRS AC PWR FAILED

IRS 2 will shutdown after 5 min

can be result of
Loss of both eng driven generators

1

5

EVT NO PFD (ATT) INFO
respective A/P disengages

NNC: IRS FAULT

NO AUTOLAND - NO P-RNAV

A/P @ APPR = NOT AUTHORIZED

IF BOTH IRS FAULTY --> RELY ON
STANDBY AI AND MAGN COMPASS

ADVISE ATC, REQ VECTORS

2

1

NNC: IRS DC FAIL

NO AUTOLAND

IF BOTH DC FAIL:
POSS. DEPLETED BATTERY OR
SWITCHED HOT BATT BUS FAIL
--->
CONSIDER STARTING APU EARLY
..IF NO APU --> TIME CRITICAL

2

2

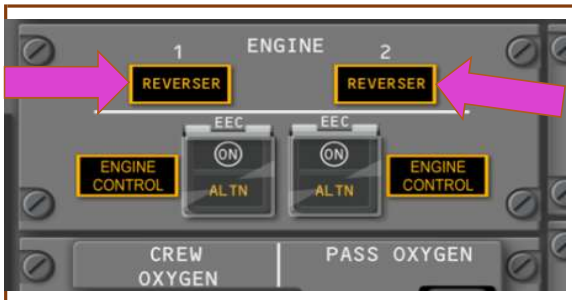
NNC: PSEU

ILLUMINATES ON GROUND ONLY

IF DURING DISEMBARK:
MIND TAILTIP (STRUT = EXTENDED)

2

3



NO REV LIGHT ON UPPER DU

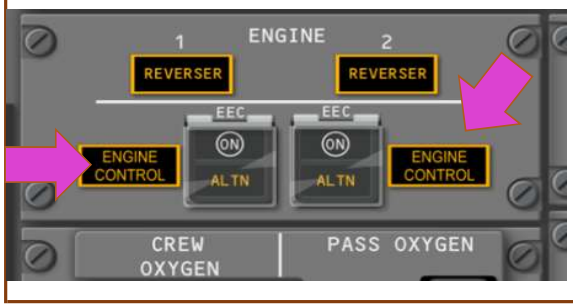
NNC: REVERSER

TO & LAND PERF

NO DISPATCH WITH BOTH REV INOP

2

4



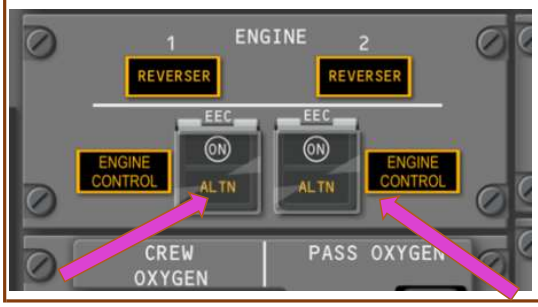
NNC: ENGINE CONTROL

ILLUMINATES ON GROUND ONLY
BELOW 80KTS

DO NOT TAKEOFF

2

5



NNC: EEC ALTERNATE MODE

possibly also DSPLY source, do this 1st TO PERF

If both EEC's altn --> probably DEU failure)

- DEU gives info to CDS (displays) and the EEC
- In EEC ALTN mode, EEC reverts to simple thrust schedule w/o limit protection. Manually monitor N1 to prevent eng overboost as rated thrust can be reached before full thrust lever travel

3

1

NNC: FLIGHT RECORDER OFF

continue normal flight

3

2

NNC: PASSENGER OXYGEN ON

check pressurization
(other NNC or memory items applic?)

descent to
FL250 (if normal pressurization)
or if necessary: 10.000ft or MSA
because OX supply only 12min

3

3

NNC: STANDBY HYDRAULIC LOW QUANTITY

CONTINUE (normal hydraulics A&B)

(Qty is below 50%)

If syst B is appr 72%, than QTY of B
is below balance line.

3

4

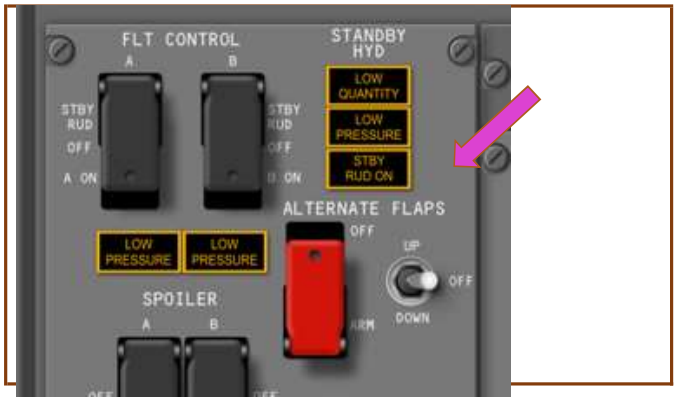
NNC: STANDBY HYDRAULIC
LOW PRESSURE

SUB: IF LOSS syst A&B: Rudder inop
LOSS A/B: respective reverser inop
LOSS B: no more

(light only armed if
STBY pump (STDBY RUD)
is activated)

3

5



NNC: STANDBY RUDDER ON

purely for awareness

if illuminated without crew action or
hydr system malfunction:
FFM detects force fight in
main rudder PCU

4

1

NNC: FLIGHT CONTROL LOW PRESSURE

Check if HYDR FAILURE is applicable

4

2

NNC: YAW DAMPER

CAT 2/3 still approved
LAND FL30 only

avoid M/S turbulence

4

3

NNC: FEEL DIFFERENTIAL PRESSURE

NO AUTOLAND (no abort needed)

mind possible A/P disengage during pitch changes

4

4

NNC: SPEED TRIM FAIL

may illuminate after
Runaway Stabilizer
-or-
after engine failure (helps to move Start switch to
CONT) to hint the EEC

4

5

NNC: MACH TRIM FAIL

Limit airspeed to 280 kts/ M0.82.

5

1

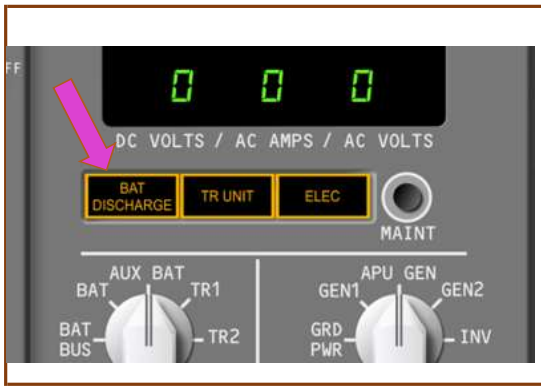
NNC: AUTO SLAT FAIL

no more stall protection
(LE slats to full ext)
during FL 1 - 5, or FL 1 - 25 on SFP ac

LE FLAPS transit light is now inhibited

5

2



NNC: BATTERY DISCHARGE

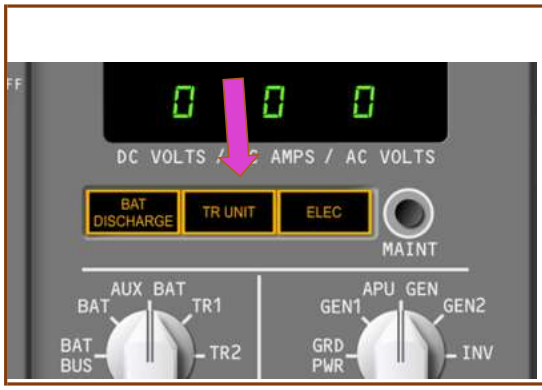
start timer (2x 30min = 60min)

Underlying problem?
Thermal runaway? Consider diversion

SUB: IF LOSS OF BOTH ENG DRIVEN GENERATORS -->
CONSIDER STARTING APU EARLY
..IF NO APU --> TIME CRITICAL

5

3



NNC: TR UNIT

LND PERF - NO AUTOLAND / CAT 3

Ground: DO NOT TAKEOFF

ILS raw data is possible to CAT 1 minima

NOGO for next flight,
consider maintenance field

5

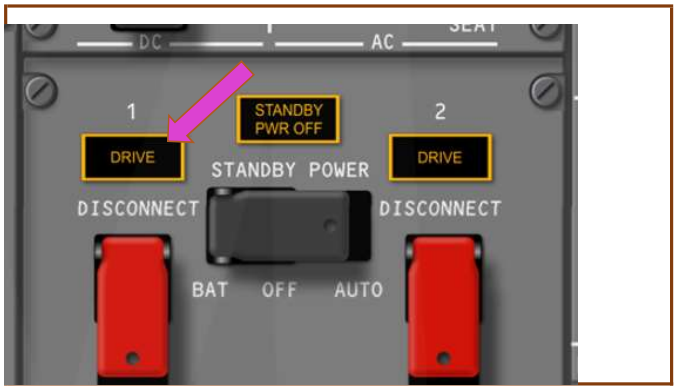
4

NNC: ELEC

Ground: DO NOT TAKEOFF

5

5



NNC: DRIVE

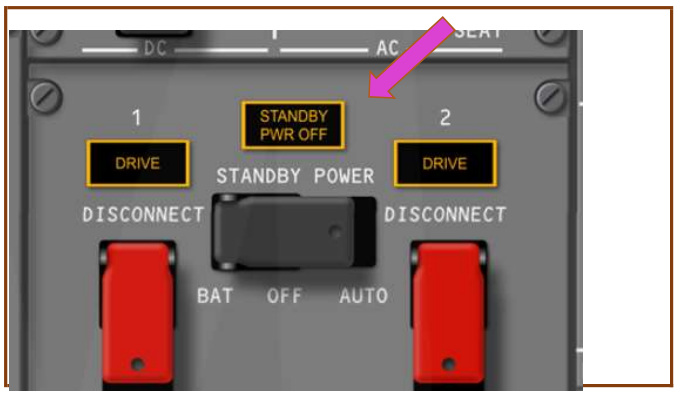
(do NNC only when ENG is not OFF)

NO AUTOLAND / CAT 3

SUB: NO APU = ONLY 1 electr source =
Plan to land @ nearest

if also TRANSFER BUS OFF and
STBY PWR OFF, then do first DRIVE NNC

6
1



NNC: STANDBY POWER OFF

no master caution illum!

One or more essential busses are not energized:
(ac stdby / dc stdby / batt bus)

LND PERF - NO AUTOLAND - NO RNP

NNC: when selecting switch to Batt ->
Forces the system to feed the stdby busses
via the batteries.

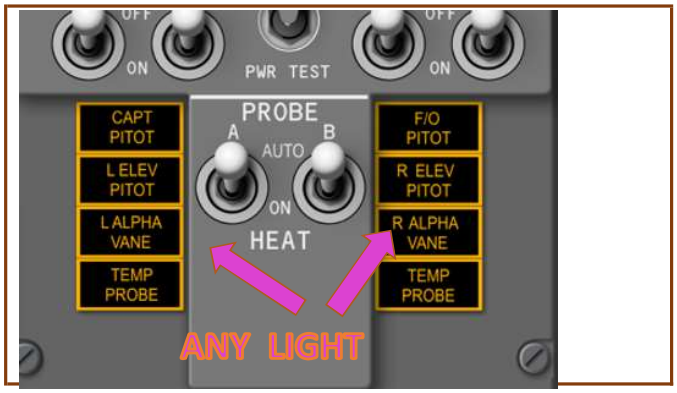
6
2

NNC: WINDOW OVERHEAT

continue

if persists:
Limit to 250kts and max 10.000ft

6
3



NNC: PROBE HEAT

AVOID ICING CONDITIONS

-may result in erroneous
flight instrument indications

possible followup failures:
- AIRSPEED UNRELIABLE
- EEC ALTERNATE MODE

6
4

NNC: ENGINE COWL ANTI-ICE

(pressure in duct is to high >65psi)

(occurs typically in the climb)
CONTINUE

evt CLB-2 setting, or V/S max 1000fpm

switch off and on resets condition
(do only when clear of icing conditions)

6
5

NNC: WING ANTI-ICE VALVE OPEN

CONTINUE

(GND: light may cycle closed/open)

7
1

NNC: ENGINE COWL VALVE OPEN
OR TAI INDICATION

@GND low idle: Can be normal -> verify setting (SP)
- APU BLD OFF and ISOL VALVE in AUTO (to avoid dual bleed)
  -> then incr thrust slightly up to max of 30% N1
    - If blue "cowl valve open" light is now dim -> all is oke.
    - If light is still bright -> valve is u/s

7
2

STEADY AMBER

NNC: ENGINE COWL VALVE OPEN
OR TAI INDICATION

@GND low idle: Can be normal -> verify setting (SP)
- APU BLD OFF and ISOL VALVE in AUTO (to avoid dual bleed)
  -> then incr thrust slightly up to max of 30% N1
    - If blue "cowl valve open" light is now dim -> all is oke.
    - If light is still bright -> valve is u/s

7
3

NNC: ZONE TEMP

if also PACK light --> first NNC PACK

(maintenance needed)

7
4

NNC: DUAL BLEED

in flight when ENG no1 thrust
can be used when:

-Eng no1 bleed air witch OFF
- Isolation valve CLOSED
-DUAL BLEED light is extinguished

7
5

STEADY AMBER +
UNCOMMANDED REV THRUST

NNC: REVERSER UNLOCKED (IN FLIGHT)

NNC QUICKLY

IF YAW IS TO EXTREME...
SHUTDOWN ENG BY HEART

IN FLIGHT & GR DN:
NO LEVEL FLIGHT POSSIBLE WITH
MAX CONT THRUST

8
1

NNC: PACK 2 PACKS = DESCENT
@APPR ---> CONTINUE
@CLIMB and 1 PACK lights ---> continue
@CLIMB and 2 PACK lights ---> lvl off
@Ground on recall: May dispatch acc Mel
SUB: PACK opposite / ENG FAIL opposite
/ Wing anti-ice valve open opposite /
Wing-body overheat opposite /
Bleed trip off opposite
DDP: 1 pack= max FL250 /
No pack = max FL100

8
2

NNC: WING-BODY OVERHEAT
Check 'bleed air duct press indic.' to confirm leak
@GND: direct sunlight can create faulty indications,
remove all bld air srcs to confirm(should clear)
@FLT: CONTINUE
If wing anti-ice is needed, consider shortly BLD air switch ON, accepting overheat, followed by turning OFF

SUB: PACK opp. side: Pressuriz. is lost! See flashcard
ENG fail opp side: Pressuriz. is lost! Descent 10000ft, or 17000' if overheat is RH side (APU can be used LH side)
and/or temperary accepting overheat-> SW bld air ON

8
3

NNC: BLEED TRIP OFF *(overpressure or overheat)*
can happen if Wing AI is on > FL350; avoid icing conds
SUB:
- 'ENG FAIL oppos'. or 'BLD TRIP OFF oppos'. side -->
Press is lost --> High spd Descent FL100 / FL170 (APU)
- PACK oppos side: All PACKS are now OFF by NNC,
turn working PACK ON (and ensure open ISOL VALVE)
- Wing-body overheat oppos. side: All PACKS are now
OFF by NNC, no workarnd, prior switching PACK OFF ->
advise ATC PAN+ imm descent FL100/FL170 (apu)
- WING AI VLVE open oppos. side: Wait with NNC,
first descnt w idle thust (acceptble blue lght) to FL140 > NNC

8
4

NNC: AUTO FAIL OR
UNSCHEDULED PRESSURIZATION CHANGE

NNC: moving switch to ALTN extinghs. AUTO FAIL light

IF NO BLD T/O: First restore configuration, then NNC
IF T/O: Climb max FL 100 (pressurization problem)

8
5

NNC: OFF SCHEDULE DESCENT

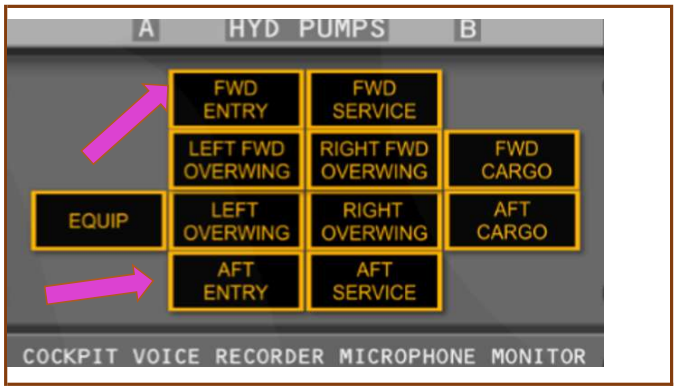
If proceeding back to DEPARTURE airport,
then no action is required

9
1

NNC: CABIN ALTITUDE WARNING OR RAPID DEPRESSURIZATION

1st headset & dont forget switch to MASK, -->
and cancel the warning ALT HORN CUTOUT
Cabin alt. controllable: also check wether it is
controllable with advancing thrust levers.
When @ safe alt: (Pilots masks off & Masks-off call), recap/feelings FO,
Ask purser sitrep (damage/injuries)
After making plan, ill give you briefing.
Make smooth descent < 10000ft --> *Deferred item:*
@pattern ALT: outfl.vlve move to OPEN untill fully open

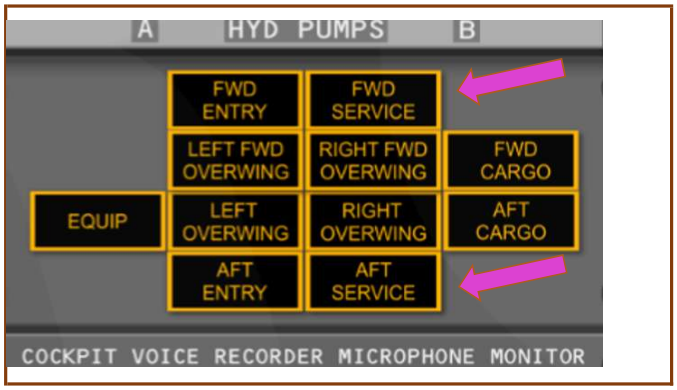
9
2



NNC: ENTRY DOOR

If pressurization looks normal & handle is closed -
-> continue normal
else --> Land at the nearest suitable airport

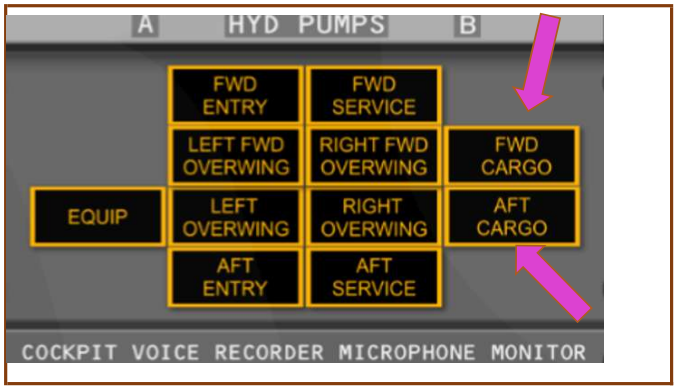
9
3



NNC: SERVICE DOOR

If pressurization looks normal & handle is closed -
-> continue normal
else --> Land at the nearest suitable airport

9
4



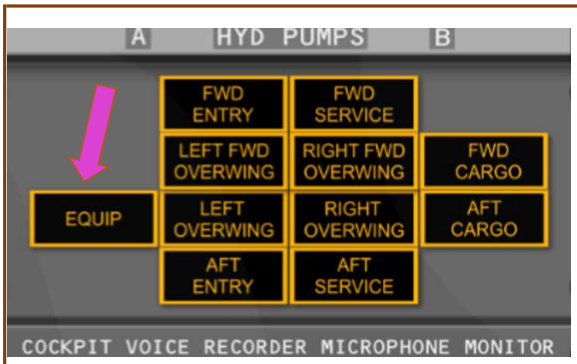
NNC: CARGO DOOR

@CLB: do not continue climb

During NNC the CABIN ALT WARNING can occur.
Common sense-> prepare&brf, and do only extra items:
-PASS SIGNS ON, PASS OX SW ON. + horn cutout.

If pressurization is abnormal (so not spurious)
--> objective of NNC is to obtain zero diff. press.
between lower&upper deck

9
5



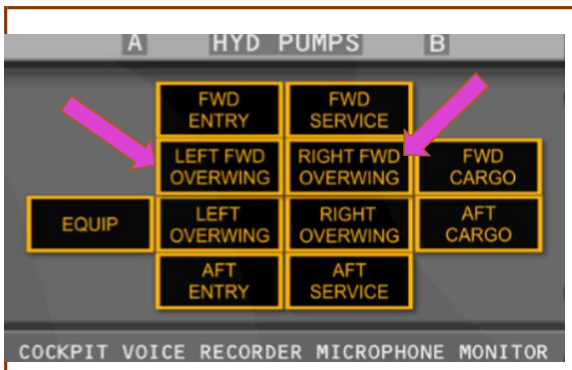
NNC: EQUIPMENT DOOR

@CLB: do not continue climb

During NNC the CABIN ALT WARNING can occur.
Common sense > prep.&brf, do only extra items:
-PASS SIGNS ON, PASS OX SW ON. + horn cutout.

If pressurization is abnormal (so not spurious)
--> objective of NNC is to obtain zero
diff. press. between lower&upper deck

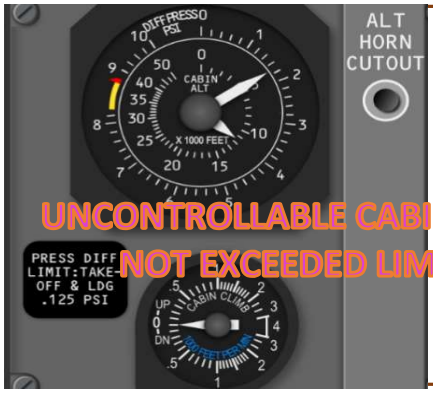
10
1



NNC: OVERWING DOOR

If pressurization looks normal & handle is closed
--> continue normal
else --> Land at the nearest suitable airport

10
2



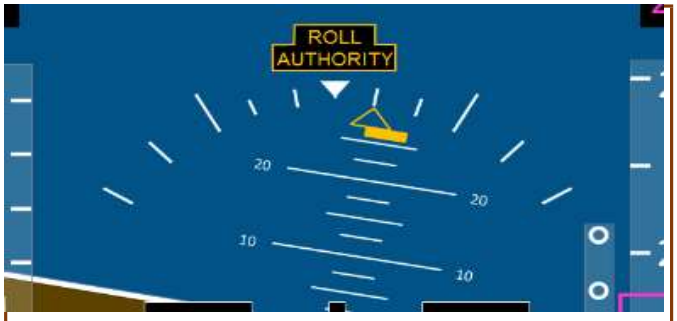
NNC: AUTO FAIL OR
UNSCHEDULED PRESSURIZATION CHANGE

NNC:moving switch to ALTN extinghs. AUTO FAIL light

IF NO BLD T/O: First restore configuration, then NNC
IF T/O: Climb max FL 100 (due pressurization problem)

Deferred item: @ pattern ALT:
outfl.vlve move to OPEN untill fully open

10
3



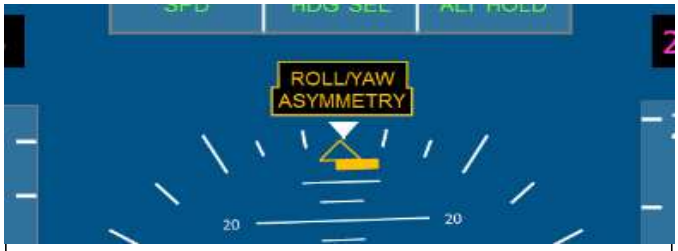
NNC: ROLL AUTHORITY

Authority limit is Reached

Check asymmetry in a/c:
trims, thrust, flaps,
icing, jammed/restricted cntrls

1st: Roll/yaw asymm.= A/P reached 75% of its authority
2nd: Roll authority = A/P reached 100% of its authority

10
4



NNC: ROLL/YAW ASYMMETRY

Authority limit is approaching

Check asymmetry in a/c:
trims, thrust, flaps,
icing, jammed/restricted cntrls

1st: Roll/yaw asymm.= A/P reached 75% of its authority
2nd: Roll authority = A/P reached 100% of its authority

10
5

NNC: EMERGENCY DESCENT

(or AUTO FAIL or UNSCHEDULED PRESSURIZ. CHANGE)

PA: "2x EMERGENCY DESCENT"
SQ 7700 (ATC), evt MAYDAY to clear airspace
LVL CHG -> Check lateral and 10000ft/MSA (fly a/c)

Limit IAS (appr. 250kts) when:
Open door/airframe vibr/struct damage
@lvl off: Maintain high spd if still above high terrain

When @ safe alt *(9000')* : *(Pilots masks off 1v1 & Masks-off call),* LTP:
recap/feelings FO > Ask purser sitrep (damage/injuries)
After making plan, ill give you briefing.

@ patt alt: outflow valve to OPEN

11
1

dual engine failure
not in gliding range of land

NNC: DITCHING

FL 40 LND

(uncontrolled fire / fuel low / fuel contamination
dual eng fail / struct damage)

Ditching with eng pwr improves a/c control
NNC will guide you. SQ 7700

"Prepare for ditching" + WWW
@2500ft: "ditching is imminent"
@500'ft "brace for impact"
300fpm descent, touchdn @ 12˚ pitch, idle after touchdn

11
2

receiving "BOMB THREAT"
message from ATC
(AIRBORNE)

NNC: BOMB INCIDENT

(only when threat is RED or AMBER?)

Plan to land at nearest suitable airport
@CLB/Descent: LEVEL OFF
Direct purser waarschuwen
Time for bomb search (45-60min)

LNDG CONFIG before descending
below actual cabin ALTITUDE

11
3

NNC: TAIL STRIKE

(obj: equalize pressure to prevent further damage)

Land at nearest suitable airport (NNC)

Takeoff = 11˚
Landing = 9˚ (due to compressed struts)

Symptoms: - a noticeable bump or jolt
- a scraping noise from the tail of the airplane
- pitch rate stopping momentarily

11
4

NNC: WINDOW DAMAGE

- FORWARD (L1,L2,R1,R2)
- HEATED SIDE (L3, R3)

(window arcing (electr) is also damage)

@ climb: stop climb if on inner plane

If no vision -> autoland

If inner plane: NNC > LANSA (below 250kts)

11
5

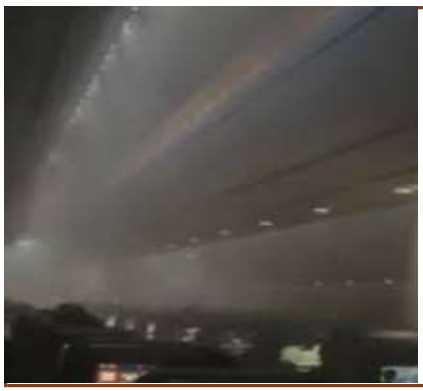
DURING T/O ROLL

NNC: WINDOW OPEN

@ T/O: Maintain maneuver speed
for existing FLAP setting -->
close window --> NNC

Avoid increasing speed.
More speed = more difficult to close window

12
1



NNC: SMOKE, FIRE OR FUMES
(Go to '*Smoke or fumes removal' NNC*,
when smoke or fumes become the greatest threat)

Ask purser if cause is known. If not certain fire is out -> divert (STP).
Keep cockpit door closed
If needed: Oxygen masks (pilots only!)
Emergency landing ops.
If no time (imm lnd): 2xUtils off, 2xEquip ALT, 2xRecirc off, when <3000ft -> even Packs off

12
2

NNC: CABIN TEMPERATURE HOT

@ GND: stop taxi
@ T/O: max FL 100
@ crz: -> NNC guides you

12
3

- LIQUID WATER ON WINDSHIELD AT TEMP TO COLD FOR RAIN (NO AIRFRAME ICING)
- L / M TURBULENCE
- ST ELMO'S FIRE (SMELL OZONE)
- CONVECTIVE WX / CLOUDS
- AMBER/RED RDR RETURNS BELOW A/C OR GREEN IN FRONT
- >30000FT ALTITUDE

NNC: ICE CRYSTAL ICING

obj: to exit ICI-conditions and reduce it's effects

Go out of the conditions ASAP
Eng AA - ON

12
4

No response .
thrust lever .

NNC: ENGINE LIMIT OR SURGE OR STALL

@ clb: Stop clb FL200 (due poss ENG restart)
@appr: if eng conditions stabilized after
Mem.Items -> consider continue appr

Normal EGT is roughly N1 x8-10. Idle is 500-600.

In some conditions, imm. eng s/dw may be necessary to maintain directional control.

12
5

receiving "BOMB THREAT" message
(ON GROUND)

NNC: BOMB INCIDENT

NNC: Disembark passengers and crew with all cabin baggage.
(by steps / jetties or evacuation)

Nobody is allowed to enter A/C until search is completed.
Isolate all passengers including cabin baggage.

13
1

N1/N2 does not incr, or very slowly after EGT increases
-or-
No oil psi indication when engine is @ idle
-or-
EGT does not incr by 15sec after engine start lever is placed in idle
-or-
EGT quickly nears/exceeds the start limit

NNC: ABORTED ENGINE START

Continue pushback --> Quickly NNC,
ENG needs motoring for 60sec

max motoring 1% within 5 sec
Starter usage max 2min.
10 sec between start attempts

If small fuel leak after ENG start from the ENG drains --> Continue.
Let ENG run for upto 5min, if leaks stops within this time, no action needed.

13
2

ENG indications abnormal

ENG indications approaches or exceeds red limit

Abnormal ENG noises/vibrations
(possibly after bird hit)

NNC: ENGINE LIMIT OR SURGE OR STALL

(flames does not always have to mean NNC 8.2)
Maybe this NNC leaves ENG on (or idle)

Vibr + limit exceed. = NNC 8.2 (severe damage)
@ clb: Stop clb FL200 (due poss ENG restart)
@appr: if eng conditions stabilized after Memory Items ->
...consider continue appr
Normal EGT is roughly N1 x8-10. Idle is 500-600.

In some conditions, imm. eng s/dw may be necessary to maintain directional control.

13
3

NNC: ENGINE TAILPIPE FIRE

Condition: ON GROUND! with no ENG fire WRNG
Reported by others without flightdeck indication
Typically during ENG start/shutdown

Motoring is key.. Do NNC quickly.. will guide you.

No need for EVAC, contact cabin to calm the passengers
"controllable fire, no need for evac"

ATC (via NNC): Pan call, req fire brigade

13
4

NNC: LOSS OF THRUST ON BOTH ENGINES

If low lvl <FL170, NNC is not 100% valid, use APU start assist

Cpt takes controls, MAN flt (stby power). If stable -> MEMitems
>FL270 = 275 kts | <FL270 = 300 kts | <FL170 = FL up spd (220)
start APU | 7700 Mayday call (CPT) | LANSA | outflw valve |
Low lwl: Try APU assisted start (like on the ground)
If no N1 indication -> sev damage -> dont waste much time
If any generator avlb -> turn on -> A/P on
<FL170 = FL UP SPD, is best gliding speed **(2,5NM / 1000ft)**
MCP ALT bug geeft green arc = indicatie gliding distance
You lose appr. 2000ft with 180˚ turn

13
5

N2 = not 0

NNC: ENGINE FAILURE OR SHUTDOWN

(if vibr/smoke/overheat/surge/stall --> other NNC)

ANC -> RUDDER (lower aileron side) -->
Focus on IAS, remaining eng thrust, yaw, automation (oth A/P) >
Then Communicate (PAN/MAYDAY)

@TO: Perform EOP, MAX CONT THRUST
@CRZ: ENG DRIFTDOWN PROC
@final: if stabil-> you can continue. If FL15 -> Vref = FL40+20
Go around: If FL30/40 -> FL15, if FL 15 -> FL 1

Considerations Eng restart: ca. 5min & 200kgs fuel) / risk of N-2 due fuel contam. / Wx (autoland)

NNC: ENGINE FIRE ON THE GROUND

Subgoal: prevent pax to evac (inform by NNC)

Fly the a/c:
- *Stop, park brake, "CC at your stations"*
- *When advise ATC: "TRA xx Stopping .. Mayday 3x fire"*

EVAC when indication of fire continues: use all sources (atc/handling/cabin/puschback driver)

If evac not needed:
- *"Passengers, this is CPT, please remain seated!"*
- *Cabin: inform fire is under control*

An engine start is needed (AIRBORNE)

- There was: NO eng fire
- There is: N1 rotation (N1 is not zero)
- There is: No abnormal airframe vibration

NNC: ENGINE IN-FLIGHT START

(No obligation, do if needed due to:
wx / notams / remote diversion / expecting other eng failure)
It may also be fuel contamination, or leak -> rather land ASAP

EEC abnormal start protection is inhibited in-flight!
Max EGT 725˚C (950˚C with loss of thrust on both eng's)
Start lever idle at minimum 11% N2

or / and

Change of fuel imbalance of 230KG in 30 minutes or less

NNC:FUEL LEAK ENGINE

(this does not adress a tank leak)

@eng start: if small stream and stops < 5 min -> continue

NNC: if s/d eng + low fuel (760kgs/tank) -> elec hyd pump overheat -> Loss of hydr syst A/B

LANDING WITH ONE ENG INOP

NNC: ONE ENGINE INOPERATIVE LANDING

FL 15 LND - NO AUTOLAND - LND PERF

Technique can be to perform NNC upto deferred items during NNC phase. Then long term plan.

Winst Bleeds OFF = 0.2-0.3% (GA CLIMB GRADIENT)

Calculate GA climb gradient without MEL Non-norm in DynSource, or check with PD tables.

NNC: UNABLE REQD NAV PERF - RNP

ANP is not sufficient

@appr: GA, unless suitable vis reference
Or switch to backup approach

Verify RNP settings in FMC-progress page4

15
1

NNC: CONFIG

Awareness light (pump switches)
Perform NNC in level flight only.

15
2

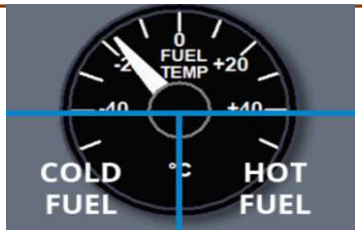
FUEL QTY INDICATION IS BLANK

NNC: FUEL QUANTITY
INDICATION INOPERATIVE
(possible VNAV disconnect)
Check if no DEU failure (DSPLY SOURCE)

Causes missing info on FMC
No approach speeds until entering GW / FUEL in FMC
PERF INIT (NNC)

No more indications of imbal / fuel leak

15
3



FUEL TEMP 'APPROACHING' :
3 DEGR AVOVE FUEL FREEZE POINT
-OR-
MINUS 43 DEGREES

NNC: FUEL TEMPERATURE LOW

(JET A1 freezing point is -47 ˚ C)

- Keep an eye on it yourself; there's no warning
- Can occur during uplift of cold fuel icw long flight

15
4

NNC: LOW

Less 453kgs in related main tank
appr. 200 kgs per 5min flight -
Maintain Clean config as long as possible
Avoid high pitch att, fly smoothly
If no time for this NNC:
Crossfd open -> all fuel pump sw - ON

Mayday 3x Fuel -> emergency landing
Minimum fuel for GND! ops electrc motor-driven pump = 760kgs -> risk OVERHEAT

15
5

NNC: HYDRAULIC PUMP OVERHEAT

Minimum fuel for GND! ops electrc motor-driven pump = 760kgs -> risk OVERHEAT

NNC: If hydraulic press > 3300psi..
-> Results in LOSS OF SYSTEM X

16

1

NNC: LOSS OF SYSTEM A *(if ENG#1 is OFF, EDMP light can stay out)*

You can Engage A/P B! LAND PERF - NO AUTOLAND

@taxi: if needed switch to ALT nosewheel steering
@T/O & GR DN: Stop climb @MSA > Land @ T/O ALT
@T/O & GR UP: Continue climb
@Appr: Abort
@final and FL 30/40: Only continue if time critical and long RWY (+MAN GEAR EXT)

Fuel penalty when GA +50% -> likely to divert due ALTN fuel
INOP: GND spoilers | Partial FLT spoilers | No A/P A | LDG GR
Close door after MAN GEAR EXT (for alt nosewheel steering)
Plan long final due to MAN GEAR EXT
PI Flaps 30/ Dry/ max manual/ 65000kg 1210m
PI Flaps 30/ BA Medium/ Max manual/ 65000kg 2240m

16

2

NNC: LOSS OF SYSTEM B (if ENG#2 is OFF, EDMP light can stay out)

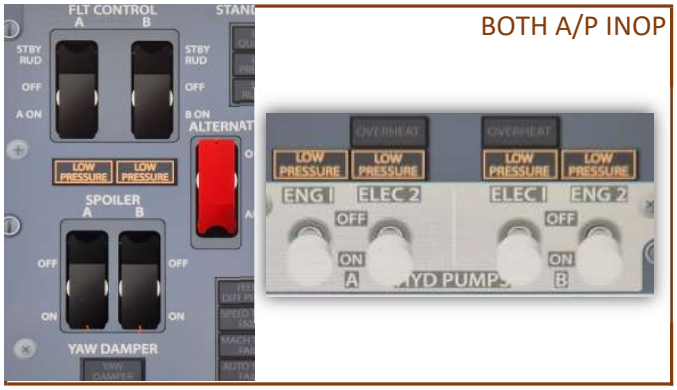
You can Engage A/P A! LAND PERF - NO AUTOLAND - FL15 LND

@T/O & LE dev ext: Stop climb @FL200, max 230kts (+10% fuel)
@T/O & all flaps up: Continue climb
@Appr: if not time critical -> abort en prepare
@final and FL 30/40: Only continue if long RWY (man. braking)

INOP: TE flaps | LE devices | YAW damper| Part.FLT spoiler | No A/P B | autobrake |
Fuel penalty +10% due to LE devices can't be retracted after ALT FLAPS EXT
ALT Flaps takes appr. 2 min (+7NM) - check assym (no protection)
-15NM line-up needed
-PI Flaps 15/ Dry/ max manual/ 65000kg 1290m
-PI Flaps 15/ BA Medium/ Max manual/ 65000kg 2365m

16

3

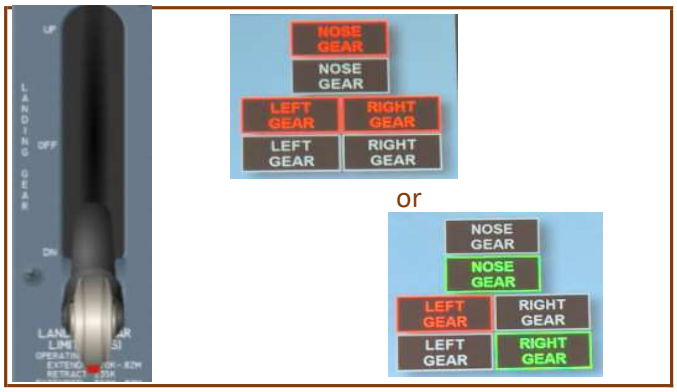


NNC: MANUAL REVERSION OR LOSS OF SYSTEM A AND SYSTEM B
(if ENG is out, respective EDMP light can stay out)

Both A/P inop LAND PERF - NO AUTOLAND - FL15 LND
Fly: Land at nearest suitable airport
If Loss of HYD syst A & B, and airborne with FL ext, or >60kts wheelspd, the Stby Hyd pump engages automatically (STBY RUD ON)
INOP: A/P A+B | TE flaps | LE devices | LDG GR | YAW damper| autobrake | | brakes (only accum. pressure left) | FLT+GND spoilers | NSwh steering
Fuel penalty: +60% due no retration of Gear and LE Flaps
- NO nosewheel steering! - Avoid wet RWY with crosswind
-below 60kts use diffrnt. braking - Towing asstn. after lndg
SUB: If Loss of STBY HYD system: Rudder control is lost, and no REVersers, and no LE Flaps > LE FL TRANSIT LIGHT (FL15+15kts)
PI Manual Reversion, Flaps 15, Dry runway LW 65T = 1720m.
Flaps 15, BA Medium 65T =3185m

16

4



NNC: MANUAL GEAR EXTENSION

Cond: - ANY LDG GEAR is not down&locked when lever is down
- LDG GR lever is jammed in OFF position

@ APPR GR LVR JAMMED OFF: abort -> NNC
@ APPR GR LVR DN: if critical, do by heart MAN GR EXT. If succesful> Land!

If lvr is in OFF position --> No nosewheel steering available

Fuel penalty (+50%). ALTernate fuel is affected. If in bad wx conditions, consider divertion prior GR extention to good weather conditions.

Close the door after use!

16

5

NNC: EQUIPMENT COOLING OFF

'Can' be indication of a pressurization problem

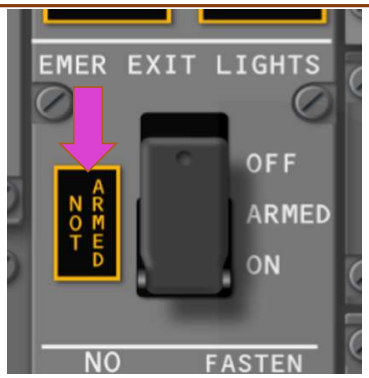
@ appr/final: continue (consider transfer cntrls?)
consider switch to ALTN if time critical

---

30 min off: can cause loss of
CAPT DU and LOWER DU (SUPPLY) or
F/O and UPPER CENTER DU (EXHAUST)

If remains on -> divert due to potential loss displays

17
1



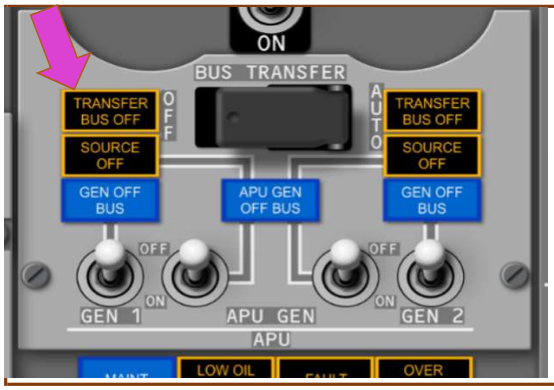
NNC: EMERGENCY EXIT LIGHTS NOT ARMED

Continue

Indication that the switch is in OFF.

If the system is activated (so switch is set to ON), then batteries provide light for 10 minutes

17
2



NNC: TRANSFER BUS OFF
(multiple lights will illum, also SOURCE OFF)
NO ILS / NO AUTOLAND / avoid ICING / FL 15 LNDG (if #2)

Short: other A/P is avail.
pan call - verlies van redundancy
Takes a lot of time, all amber lights NNC must be perf.

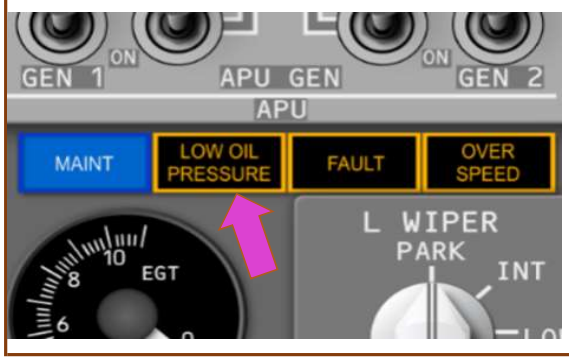
LNAV/VNAV approach is avail
(ils bars retract @ G/S capture)

17
3

NNC: SOURCE OFF
(if also 'DRIVE' / 'Transfer bus off' light illums, go to resp. NNC)
(if both SOURCE OFF lights -> LOSS OF BOTH ENG DRVN GENs)

Short: continue
- Other generator will feed the xfer bus. If no APU -> only 1 AC source remains -> Plan to land at nearest suitable airport.
- When 1 source remains: APP mode data is removed from the respective PFD upon GS capture

17
4

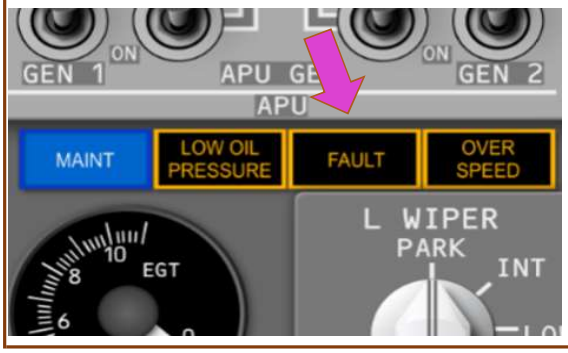


NNC: APU LOW OIL PRESSURE

System has auto-shutdown, light turn off 5min after shutdown

dispatch without APU -->
cannot perform 'nobleed takeoff'

17
5



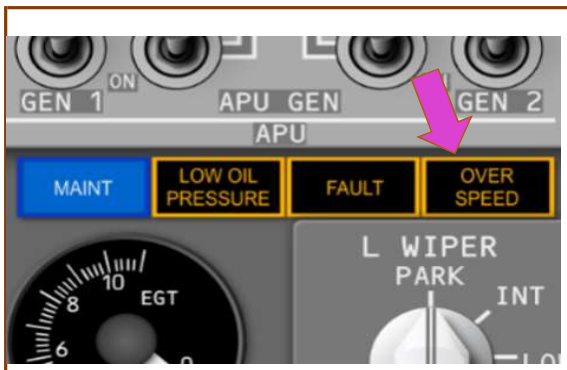
NNC: APU FAULT

The ECU self-tests 5min after selecting APU switch to OFF. If fault is no longer detected, the light extinghuish and a restart can be attempted.

(on GND: within 5min ECU prevents start attempts.)

dispatch without APU -->
cannot perform 'nobleed takeoff'

18
1



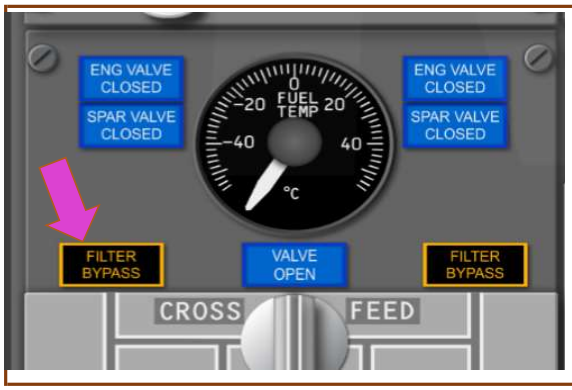
NNC: APU OVERSPEED

System has auto-shutdown, light turn off 5min after shutdown

An aborted APU start can trigger 'overspeed' light. A 2nd start is allowed without maintenance.

dispatch without APU -->
cannot perform 'nobleed takeoff'

18
2



NNC: FUEL FILTER BYPASS

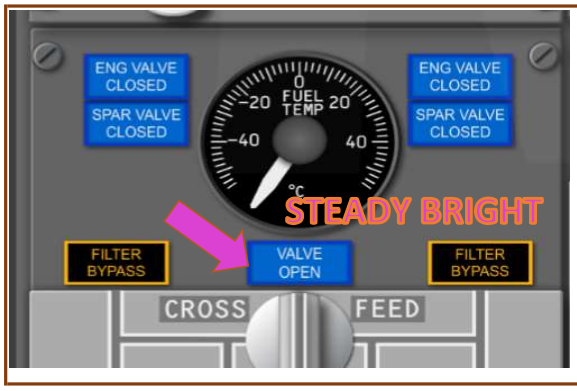
(= impending or actual bypass)

STP: 1 light -> continue
2lights -> divert immediately. (emerg landing)
If then 1 eng fails -> consider landing immediately (consider off field).

SUB: 2 lights and N-1 on same side = unusable fuel on that side
2 lights and N-1 on other side = all fuel is usable

If both lights: consider switch off center tank fuel pums, fuel in main tanks is less contaminated, since it is mixed with remaining fuel from last flight, AND land ASAP.

18
3



NNC: CROSSFEED SELECTOR INOPERATIVE

(only noticable during switching, no MC)

STP: continue

Worst case, no exchange fuel left-right. So imbal may grow. Consider use of assym thrust.

Check CB, if popped -> no light/indication anymore, so unknown position. Do not perform 'NNC IMBAL'!

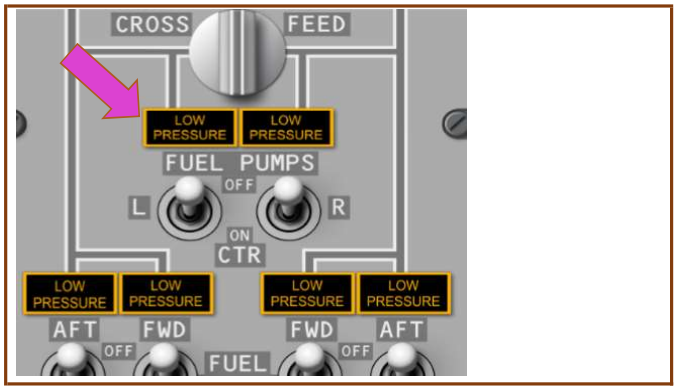
18
4

NNC: ENGINE FIRE OR ENGINE SEVERE DAMAGE OR SEPARATION

FLY A/C: ANC -> RUDDER (lower aileron side) -->
Focus on IAS, remaining eng thrust, yaw, automation (oth A/P) >
Then Communicate (PAN/MAYDAY)

@gnd: --> ENG FIRE ON THE GROUND (backcover)
@TO: --> 12-13gr pitch, Perform EOP, (MCT), MEM items
@CRZ: -> analyse -> FLY ENG DRIFTD PROC (ANC), MEM items
@final: if stabil-> you can continue. If FL15 -> Vref = FL40+20
Go around: If FL30/40 -> FL15, if FL 15 -> FL 1

18
5



NNC: FUEL PUMP LOW PRESSURE

(lights may flicker in turbulence, or climb/descent & low)

Auto shutoff when 15sec lights on

1 light -> continue

2 lights same side -> consider descent FL300/250 to improve suction feed and avoid flameoout

2 light center tank = cntr-tank fuel is unusable (calculate) also add the unusable fuel to the final reserve (FMC calc)

Lights icw fuel filter = possible heavy contamination

19

1

+ ENGINE FIRE SWITCH ILLUM.

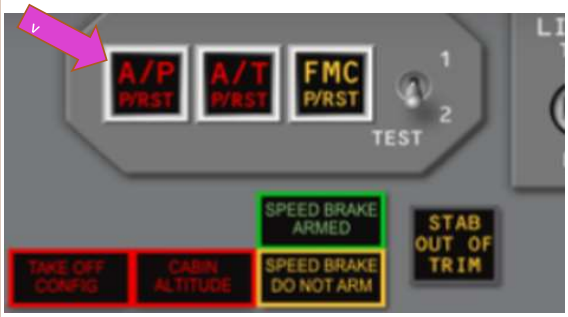
NNC: ENGINE FIRE OR ENGINE SEVERE DAMAGE OR SEPARATION

FLY A/C: ANC -> RUDDER (lower aileron side) -->
Focus on IAS, remaining eng thrust, yaw, automation (oth A/P) >
Then Communicate (PAN/MAYDAY)

@gnd: --> ENG FIRE ON THE GROUND (backcover)
@TO: --> 12-13gr pitch, Perform EOP, (MCT), MEM items
@CRZ: -> analyse -> FLY ENG DRIFTD PROC (ANC), MEM items
@final: if stabil-> you can continue. If FL15 -> Vref = FL40+20
Go around: If FL30/40 -> FL15, if FL 15 -> FL 1

19

2



NNC: AUTOPILOT DISENGAGE .

NO AUTOLAND (no RNP AR)

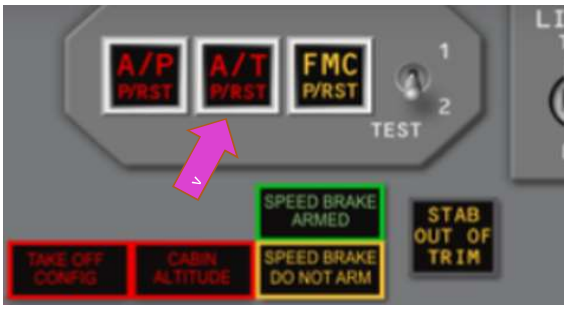
STP: fly manual or re-engage

DDP: 3 hours fight w/o A/P is considered acceptable, longer is SCD

Possible cause is FCC fail (no NNC), Now also FD flag.
If dual FCC fail -> MACH trim fail, SPEED trim fail & SEL SPF will show.

19

3



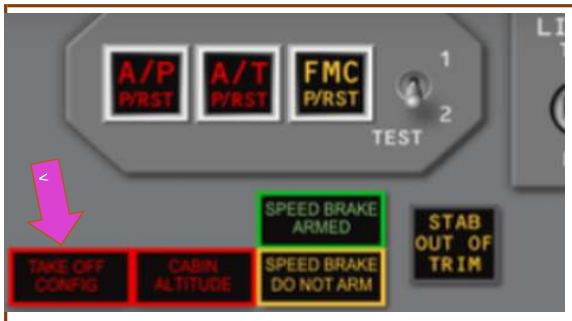
NNC: AUTOTHROTTLE DISENGAGE .

STP: manual thrust or re-engage

DDP: A/T is not required for ANY approach

19

4



+/or WARNING HORN INTERMITT.

NNC: TAKEOFF CONFIGURATION

(GND only: intermittent horn, when adv thrust levers)

Airborne = Cabin alt wrng

<80kts = RTO

>80kts, consider RTO... if unable or unsafe to fly.
Wrong config is unsafe to fly.

19

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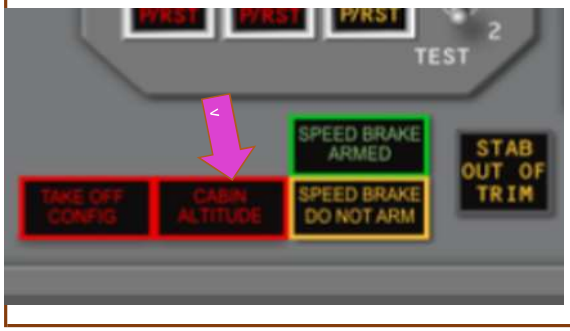
NNC: ---none---

Take immediate positive corrective action to adjust the flight path or correct the configuration until the alert ceases.

20

1



NNC: CABIN ALTITUDE WARNING OR RAPID DEPRESSURIZATION

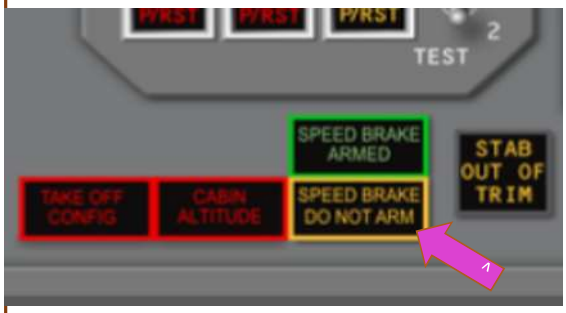
TUC: FL400 20-30s

CRZ: After masks on -> kill bell & start Mem items -->
Emer descent->lowest safe alt / 10000' (mayday/7700/QNH)
PA: Emergency descent 2x (lvl change, spd brake) check anti-ice

Then -> NNC (both items)
After NNC -> STP: check maneuvering to landing area
Ask purser how cabin&pax are doing
> 10000ft -> masks off

20

2



NNC: SPEED BRAKE DO NOT ARM

(auto-speedbrake fault, manual does work) TO/LND PERF

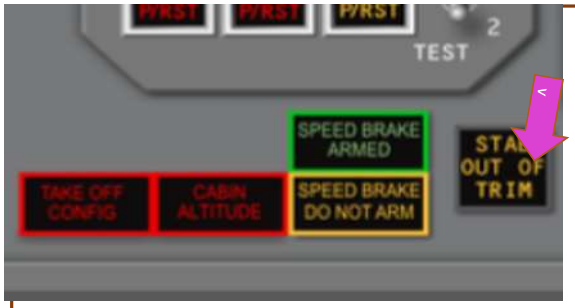
Climbout: upclean a/c -> NNC
Appr: Continue if not critical distance (+70meter),
brief MAN speedbrake deployment

Brief: Handle in DN position for landing ->
Manual deployment speedbrake and
"speed brakes NOT up" call

Dyn source: mel item 27-07 for calculation
Perf Inf:(man speedbr): norm.table notes: add 50-70m to LDG dist

20

3



NNC: STABILIZER OUT OF TRIM

(a/p does not set trim correctly)

FL15 LND - LONG FINAL - NO AUTOLAND - NO RVSM

momentary light is normal during large trim changes

- If stab is trimming (check by forcing a trim situation)-> continue
  -> working on large trim changes. (momentary light)
- If stabilizer is NOT trimming -> disengage AP -> if still not able
  manual trimming -> NNC stabilizer trim inoperative (FL15 LND)

All above is also in NNC.

@ final: when manual trim not working -> G/A due to possible
lack of pitch authority in case of Flare/Goaround

When in RVSM: "unable RVSM due to eqpmt"

20

4

LANDING with TE FLAPS
less than 1

NNC: TRAILING EDGE FLAPS UP LANDING

--> via NNC TE FL asymmetry / disagree
LND PERF - NO AUTOLAND / CAT 3
Req straight ahead MAP, due to limited bank maneuv.
Lndg Vref is = FL 40 + 40 kts (LE flaps FULL EXT)

65ton, max REV thr, max manual =
(DRY) 1400m / (GD) 1800m / (GD/MD) 2062m / (MD) 2500m

PA: high spd & firm , emerg servics to expect after lndg
Hot brakes 15min after LND, ask for fire brigade to monitor

20

5

NNC: TRAILING EDGE FLAP DISAGREE

@ Appr: abort lndg. Unless fuel limited.
If continue (rwy suff): Set new Vref + Flap inh.sw.+ a/b max
Vref: (FL up + LE FL full ext = FL40 + 40)
PAN Call, ivm degr performance MAP, custom MAP?
BRF FLOW ALT FL EXTENSION: 1) FL LVR -> 2) SPD BUG -->
3) thr reductie (only when FL are set with no asymm
ALT FL EXT --> Geen asymm protct, allow 15s rest betw. / 5min
GR warning horn, when FL > 10 & GR DN, consider "GR DN FL15"
If indicator fail: speedtape helps ID current FL setting
Consider retracting ALT FL EXT when longer flight to divert
BRF MAP: FL retr only when workload permit, so FL15 MAP!

21

1

NNC: AUTO BRAKE DISARM

LAND PERF

@GND: no takeoff (dispatch after maintanance actions)
@ T/O: only reject <80kts (brake manually!)

Brief:
- dont forget MAN braking (right after main gear t/d) !
  And PM monitor manual braking

21

2

NNC: ANTISKID INOPERATIVE

(also probably no autospeedbrake)

TO / LAND PERF - NO AUTOLAND

@GND: no T/O (disp. after maint actions, DRY rwy only)
@final critical: only long rwy, autobr switch off, speedbr
DN detent -> max man braking and MAN speedbrake
Favor FL40 lndg, due to land dist.
Manual speedbrake, MAX man braking

PI: 65t Max rev: FL15 dry 2000 GD: 2300 GD/MD: 2500
FL40 dry 1900 GD: 2100 GD/MD: 2300

21

3

NNC: LEADING EDGE FLAPS TRANSIT

(LE FL not in cmd pos - or skew - or asymmetry)
check if no Gear disagree!

LAND PERF - NO AUTOLAND - FL15 LANDING

@ t/o: expect T/O config warn after T/O
@crz: imit IAS 230kts and ALT < FL200 -
@APPR: go-around

If cold wx, maybe shot Wing AI helps

21

4

NNC: TRAILING EDGE FLAP ASYMMETRY

(you dont move FL anymore, this is your TE FL LNDG config)
LND PERF, NO AUTOLAND

@ Appr: abort lndg. Unless fuel limited.
If continue (rwy suff): Set new Vref - Flap inh.sw.- a/b max Vref:
(FL up + LE FL full ext = FL40 + 40)
PAN Call, ivm degr performance MAP, custom MAP
If Go-around with >FL25 - -> warning horn can sound. Consider lowering LDG when clear of terrain
If LNDG TE FL UP: 2500m is allways sufficient (upto GD/MD)

21

5

NNC: GEAR DISAGREE

(red gear light = gear disagrees with lever)

Speed < 235kts (retract spd limit)

@analyse: check hydr syst A.

@final and 3greens: LAND! evt wiggle lever > GPGI switch
-> GR inhibit

- Wiggle lever in his place can solve problem. Do not recycle without NNC
- Checking MANUAL GEAR DOOR is closed properly , and recycle lever, can solve problem (via NNC)

22
1

NNC: LANDING GEAR LEVER JAMMED IN THE UP POSITION

(c/l will be started when rdy to extend the gear for landing)

After T/O: Remain ivo departure/ Takeoff Alternate.

Fuel penalty (+50%). ALTernate fuel is affected. If in bad wx conditions, consider divert prior GR ext. to good wx conditions.

If lever stays up --> NO nosewheel steering!
- Avoid wet RWY with crossw. -below 60kts use diffrnt. brkng
- Towing assistance after lndg
- DONT USE ALT NOSEWH STEERING!! (GR MAY COME UP)

22
2

NNC: LANDING GEAR LEVER WILL NOT MOVE UP AFTER TAKEOFF

(NIET de override gebruiken, tenzij terrein)

(drag penalty appr 50%) - NO AUTOLAND

Fly the N-1 procedure (due performance)

No A/T & pressuriz. & winds if air.gnd sensor is u/s

If warning horn sounds... (prblm is not lever lock solenoid) --> do not use Speedbrake, GND spoilers can deploy in flight

22
3

NNC: PARTIAL OR ALL GEAR UP LANDING

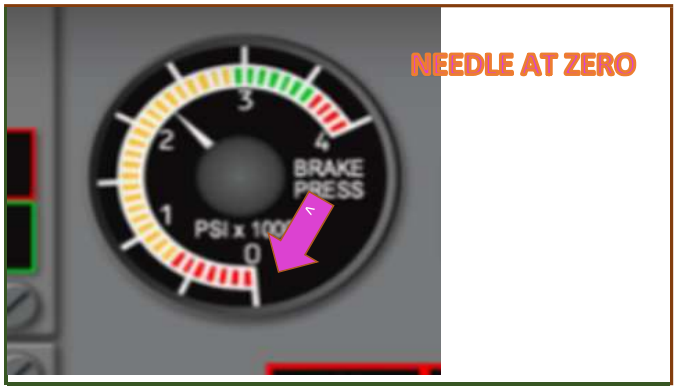
evt burnoff fuel -> slower FAS & less flammable

Actions@final: 500ft brace for impact -> ENG BLDs off -> fuel pumps off prior flare.
@LND: no speedbrake &no reverse use -> EVAC if needed

Flight deck chime n/a: Inform, en consider unlock door

OM-B: 3.15.2: techniques partial GR UP lndg

22
4



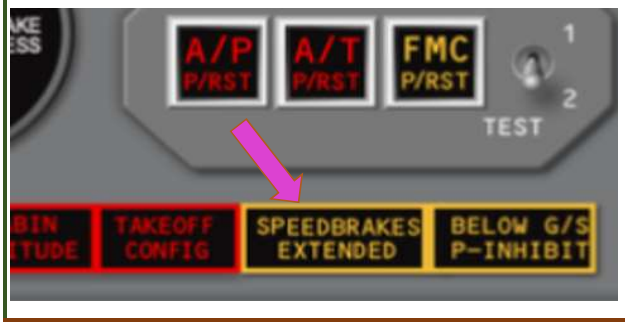
NNC: BRAKE PRESSURE INDICATOR ZERO PSI

(unannunciated: brake accum has no nitrogen precharge)

No park brake available if SYST A & B are off -> chocks

SUB: if loss of syst A&B: No backup brake pressure left

22
5



NNC: SPEEDBRAKES EXTENDED

(configuration awareness)

@GND: lever DN -> if still illum -> Do not t/o

@ airborne: only use speedbrakes when:
- more then 1000 ft AGL
- FL 10 or less

23
1

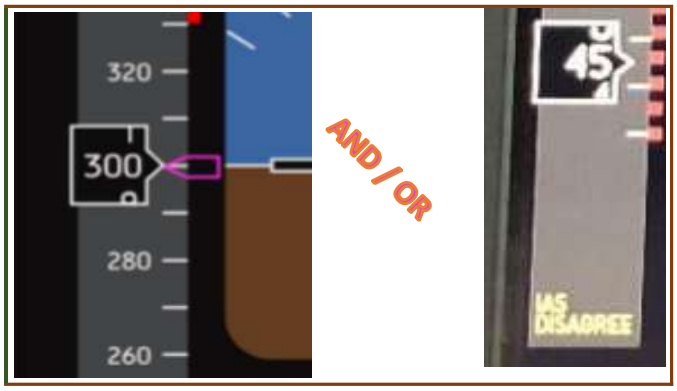
NNC: GROUND PROXIMITY GPWS INOPERATIVE

NO AUTOLAND (if mode 6 unav.=callouts)

Some or all GPWS alerts are not available.
GPWS alerts which occur are valid.

DDP: - Strictly adhere to all SOP callouts
- use autoflight to max extend as possible (especially IMC)
- all min. altitudes shall be briefed (MSA/MORA etc)

23
2

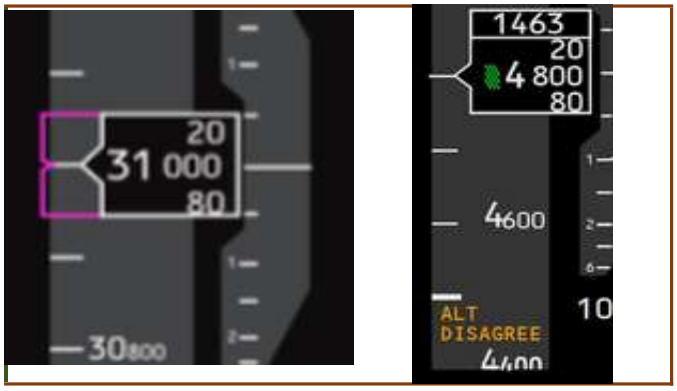


NNC: AIRSPEED UNRELIABLE

TO / LAND PERF - NO AUTOLAND - NO RVSM

@ t/o below v1: reject if unsafe to fly
@ t/o > v1: maint pitch (15˚ and t/o thrust) till accel height ->
perf MEM items pitch&thr when reaching accel height
ATC: req. airspace to perform this -> keep Pitch&thr untill NNC directs
@ arrival: abort and perform NNC -> setup long final on next try
@ short final: -if YES reliable source -> continue LNDG
-if NO rel. source & enough fuel -> MEM items & G/A
if fuel critical: continue (FL30 57% 1˚ / FL40 60% 0˚) Consider using GND spd, and higher setting autobrakes
Brief vervolg vlucht pitch&thrust settings incl G/A & lvl off
Do not use/rely on TOGA when go-around, but use PI table info
PI: pitch&thr tables: descent - terminal area - hldg area etc

23
3



NNC: ALT DISAGREE

STP: no RVSM, accept upto FL290 (check higher fuel rate)
"unable RVSM due equipment"

RVSM needs 2 primary altimeters

Use ISFD to confirm correct altimeter

Max difference: @GND =50ft (60ft >5000ft)
@airborne = 200ft

23
4

NNC: CDS FAULT

common display system

(annunciates on ground only, before 2nd eng start)

STP: Do not takeoff.
In case of "CDS MAINT" white letters, may be dispatchable after contacting maintenance.

- This is a fault in DEU's or de displays itself.
- DEU gives info to CDS (displays) and the EEC.
- (CDS supplies info on the 6 DU's)
- After 2nd eng start, it changes to "DSPLY SOURCE"

23
5

NNC: DISPLAY SOURCE

(check combo with EEC altn lights, do NNC DSPLY source 1st?)

Lost also other instr.info (speed, maneuv.bar) NO AUTOLAND

STP: - switch a/p to working DEU side
- A/T possible not optimal due to EEC ALTN

@Final/appr: continue if stable, use displays on working side

- DEU gives info to CDS (displays) and the EEC. If 1 DEU failed, remaining DEU gives info to both CDS.
- In EEC ALTN mode, EEC reverts to simple thrust schedule w/o limit protection. Manually monitor N1 to prevent eng overboost as rated thrust can be reached before full thrust lever travel

24

1

NNC: DISPLAY FAILURE

(possibly autoswitching of displays occur)

You can use MAIN PANEL DU and LOWER DU selectors

24

2

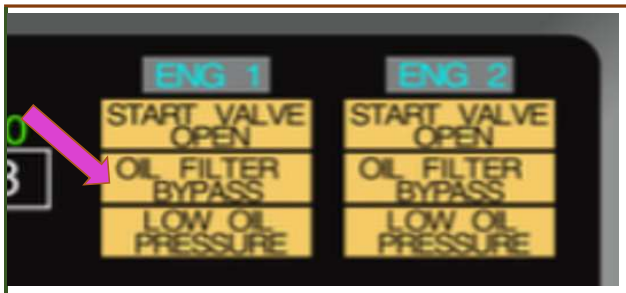
NNC: START VALVE OPEN

@ eng start: Perform NNC (no abort start!!)

@airborne: wing A/I must be switched off to prevent asymatric ice buildup

24

3



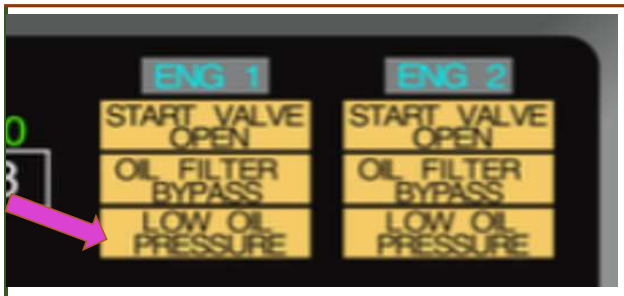
NNC: ENGINE OIL FILTER BYPASS

NNC: Reduce thrust.
If light remains -> NNC directs to ENG shutdown.
If light exting -> consider engine partially inop.
Check performance limitations
Consider NNC eng failure as guideline

-Cold wx ops on GND: light may illum (wait at idle until oil psi is normal and light must ext.)

24

4



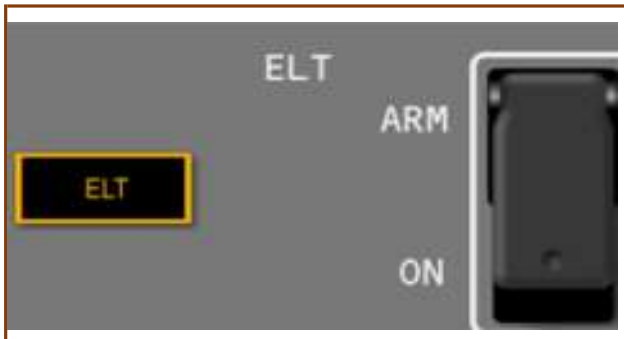
NNC: ENGINE LOW OIL PRESSURE
(icw lower DU pop-up)

NNC:
in amber band -> No takeoff
In red band -> NNC directs to ENG shutdown

-Cold wx ops on GND: light may illum (wait at idle until oil psi is normal and light must ext.)

24

5



NNC: ELT

(ELT transmitter is ON)

@GND: do not takeoff with ELT on

try reset by toggling switch to ON then ARM

25

1

NNC: IMBAL

(>453 kgs difference betw main tanks) - (long APU us?)

@ final: continue approach

check fuel flow & fuel used. If leak suspected -->

NNC FUEL LEAK ENGIINE

if IMBAL mess --> always NNC

small imbal can be perf. by SP:"fuel balancing",

but consider possible fuel leak

Flow: CROSSF open -> LOW MUST GO ->

when equal -> pumpON -> CROSSF close

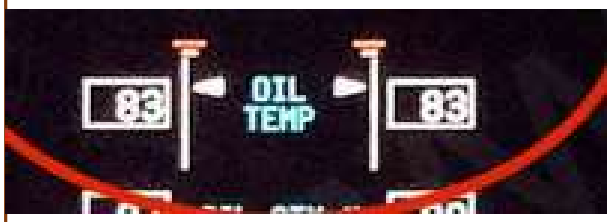
(200kgs is appr. 5min balancing)

25

2

> 140 DEGR MAX CONT FOR 45MIN

MAX ENG OIL TEMP 155 DEGR



NNC: ENGINE HIGH OIL TEMPERATURE

(icw lower DU pop-up)

STP: start timer

In red band: NNC directs to ENG shutdown

In orange band >45min: NNC directs to ENG shutdown

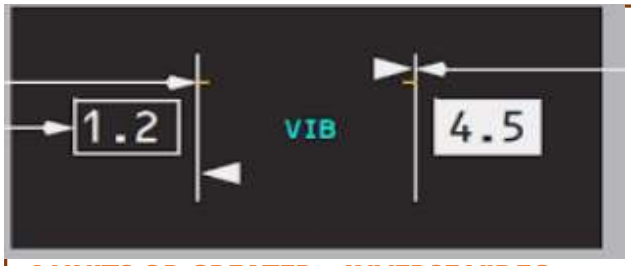
If thrust reduced -> consider engine partially inop.

Check performance limitations

Consider NNC eng failure as guideline

25

3



4 UNITS OR GREATER + INVERSE VIDEO

NNC: ENGINE HIGH VIBRATION

(icw lower DU pop-up - when 4.0 units or greater)

@GND: if > 1.0 unit with no history, consider contact maint.

Cause could be icing vs mechanical.

If mechanical: descent to lower ALT and lower IAS can lower the vibration.

@cold wx on GND: dont forget runup every 30min.

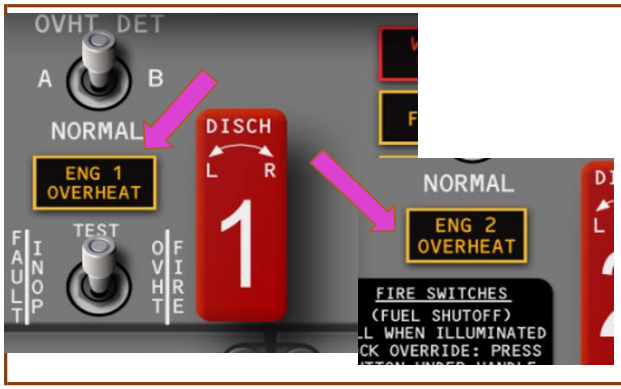
If thrust reduced -> consider eng. partially inop.

Check performance limitations

Consider NNC eng failure as guideline

25

4



NNC: ENGINE OVERHEAT

STP: consider MCT on good engine

@takeoff >80kts: continue

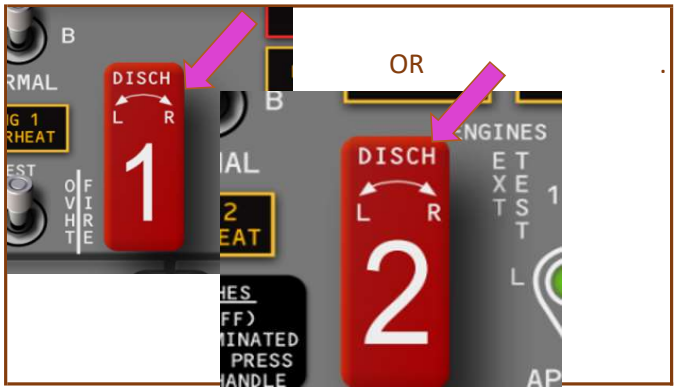
If thrust reduced -> consider engine partially inop.

Check performance limitations

Consider NNC eng failure as guideline

25

5

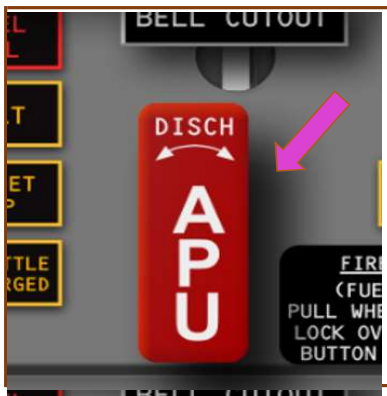


NNC: ENGINE FIRE OR ENGINE SEVERE DAMAGE OR SEPARATION

FLY A/C: ANC -> RUDDER (lower aileron side) -->
Focus on IAS, remaining eng thrust, yaw, automation (oth A/P) >
Then Communicate (PAN/MAYDAY)

@gnd: --> ENG FIRE ON THE GROUND (backcover)
@TO: --> 12-13gr pitch, Perform EOP, (MCT), MEM items
@CRZ: -> analyse -> FLY ENG DRIFTD PROC (ANC), MEM items
@final: if stabil-> you can continue. If FL15 -> Vref = FL40+20
Go around: If FL30/40 -> FL15, if FL 15 -> FL 1

26
1



NNC: APU FIRE

on GND: *APU bottle is auto-discharged > still do actions!*

@ takeoff below V1 = RTO
@ final = mem items and continue

If stays illuminated: Land at nearest suitable airport!
MAYDAY

After landing/taxi: visual confirm. fire (fire dpt) = EVAC
If in doubt, ask cabin for indications heat/smoke/smell

26
2

NNC: WHEEL WELL FIRE

fire detected in MAIN wheel well

@ takeoff below V1 = RTO
@ airborne: Spd <270K/.82M to facilitate GR retract. NNC
@ final: Mayday and continue.
Light still on after LDG: -> CC at your stations -> EVAC
Unless no indic. by TWR/CC etc. Then inspection fire dpt.
GR max extend spd (270K/.82M). Max retract spd (235K)

N=2 you have enough pwr for the Alps FL160
Fuelflow: GR DN is 50% more FF
+/- 3600 iso 2400 (@ 65T and FL250)

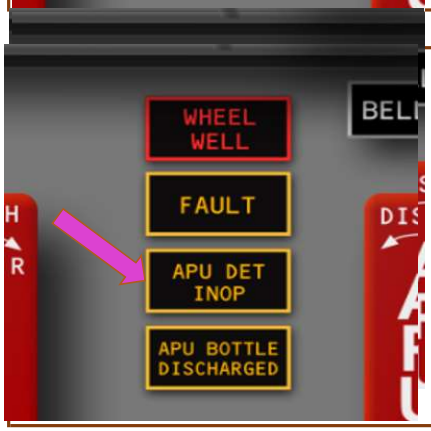
26
3

NNC: ENGINE FIRE/OVERHEAT DETECTOR FAULT

inop ENG fire and overheat detector in 1 or 2 eng's
(depending on loop selector/ OVH DET SW: A-NORM-B)
MC-light will not illuminate

@airborne: try to isolate to a working loop (suppl proc).
If no working loop left, consider diversion.
No fire warning anymore.

26
4



NNC: APU DETECTION INOPERATIVE

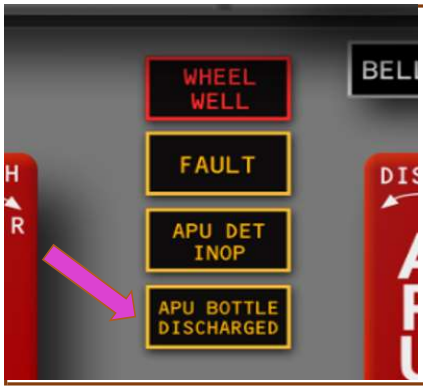
APU cannot be used anymore / shutdown APU

@airborne: mind you have no backup generator/bleed

SUB: If loss of both generators, consider using APU.
Risk management.

- Consider air starter for return flight / airconditioning
- Dispatch only with no use of APU (deactivated by maint.)

26
5



NNC: APU BOTTLE DISCHARGED

APU cannot be used anymore

@airborne: mind you have no backup generator/bleed

SUB: If loss of both generators, consider using APU.
Risk management.

- Consider air starter for return flight / airconditioning
- Dispatch only with no use of APU (deactivated by maint.)

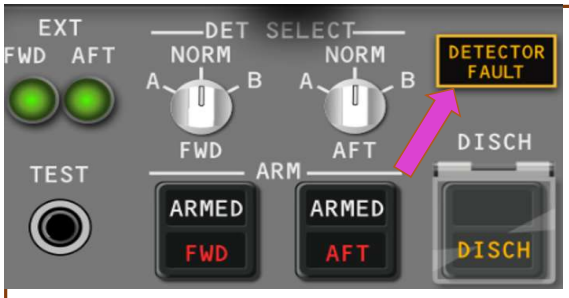
27
1

NNC: L/R BOTTLE DISCHARGED

illumes when bottle psi is below threshold

@ gnd: No dispatch allowed
@ airborne: Continue but mind 1 extinguisher less

27
2



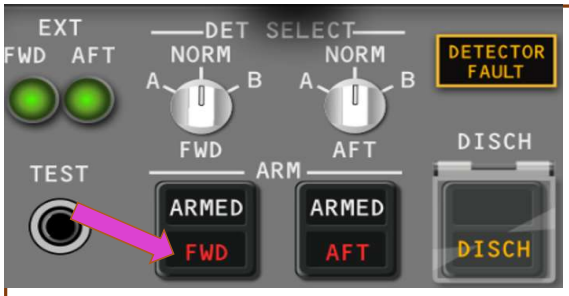
NNC: CARGO FIRE DETECTOR FAULT

One or more of the selected det. loop(s) in either cargo compartment has failed.

@airborne: Try to isolate to a working loop (how= ddp)
If no working loop left, consider diversion.
No fire warning anymore.

DDP: Dispatch allowed with 1 working loop.
If both loops fail > dispatch possible when affected Cargo compartm. remains empty. *(DDP: fwd/aft detection loops)*

27
3



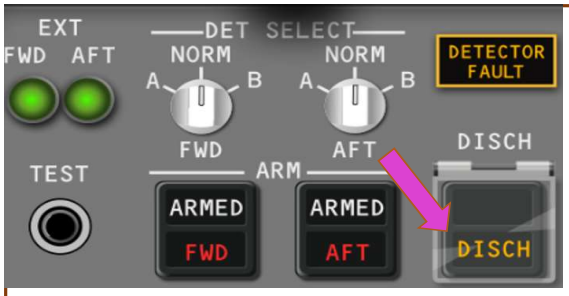
NNC: CARGO FIRE *'smoke' detected in cargo*

@ taxi: CC @ your stations
@ takeoff below V1 = RTO + "CC @ your stations"
@ airborne STP: Mayday and start diversion (time critcl)
@ final: Mayday and continue. NNC asap.

Light still on after LDG: -> CC@your stations -> EVAC on rwy
Unless no indic. by TWR/CC etc. Then inspection fire dpt.

-Upon landing: cargo doors must remain closed + inform!
-After discharge -> fire light may re-illum. Does not indicate the fire is uncontrolled.
- *Only 1 bottle available (provides 60min of fire suppression)*

27
4



NNC: DISCH

cargo fire exting.bottle has discharged or pressure is low

@ gnd: dispatch allowed if maintenance verifies the bottle has adequate charge once each flight day

@ airborne: Continue (if no fire indic) but mind no exting left

27
5

STUCK MIC

NNC: RADIO TRANSMIT CONTINUOUS (STUCK MICROPHONE SWITCH)

Try isolate faulty panel, switch this to FLT INT.

- Check handmic or headset. Consider using observer ACP.
- If nothing works, Perform lost comms procedures (LIDO)

28

1

JAMMED OR RESTRICTED IN ROLL/PITCH/YAW

NNC: JAMMED OR RESTRICTED FLIGHT CONTROLS

NO RVSM - FL15 LND - LONG FINAL - NO AUTOLAND

Hard to recognize (A/P issues). To ID jammed cntrl, apply force to each cntrl individ.(force a break-out). The one with the greatest a/c control is the non-jammed cntrl. Use this for remainder flight.

- STP: reduction in spd can help minimize unwanted movements
- Approach: use long final, custom climb out instructions
- Perform emergency landing

- Use alternative trim to aid in flight control
  (like rudder trim to aid in aileron control)
- If ice could be cause, descent to warmer conditions.

28

2

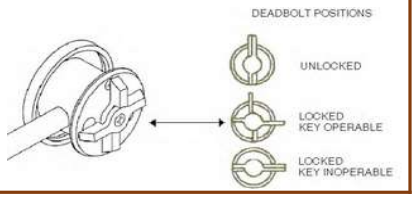
NNC: LOCK FAIL

The flightdeckdoor access switch is off or the lock has failed

Dispatch possible. (M): disables system. (O): use deadbolt proc.
Keypad not working anymore -> inform crew (airborne)

ddp: 2 crew members must be present in cockpit at all times.

Purser has key ico incap.
This can also be used
with this problem.



28

3

NNC: AUTOMATIC UNLOCK

The correct emergency acces code is entered
30 seconds response time -> then door unlockes

Switch to DENY and hold for 1 sec!!!
Only accept when legality is confirmed (CDSS).
@airborne: when not legal: start lockdown procedure.
Deadbolt procedure (clearzone if possible)

Using door-code is meant for incap situations.

28

4

AND/OR MESSAGE:
GPS L/R INVALID

NNC: ADS-B OUT FAILURE

ADS-B info= lost/degraded (possibly via GPS inv. or failed xpndr)

Indication: GPS-L or R INVALID (scratchmessage)
on selected xpndr side.
--> Select Xpndr to working GPS side.

- EASA: allmost all mode-s and rvsm required airpaces are ADS-B out mandatory.
- ADS-B: Automatic GPS-based continous broadcast of pos/alt/spd via Mode-S. Transmits every sec without interrogation. Standard xpndrs only reply when interrogated (info via gnd based rdr).

28

5

ONE OR MORE OF THESE OCCUR:

ST ELMO'S FIRE
REDUCED VIS+AMBIENT LIGHTING
SMOKE (SULFUR ODOR) / DUST
VHF/HF INTERFERENCE
BRIGHT GLOW @ INLETS / LE WING

NNC: VOLCANIC ASH

Fly the a/c: Exit asap via 180˚ turn (HDG SEL), descending (close thrust lvr's, disconn. a/t) to reduce airflow in engs.+ mayday

@STP: start diversion

If loss of thrust on both engs occur: Do memory items.
Then NNC Volc. ash prior NNC loss of thrust on both engs

If windows damaged -> consider dual channel approach. Or open side windows at final.

Sub: - IAS unreliable can occur (do NNC and use gnd spd).
- dust can trigger Cargo fire, and Equipment Cooling

Actions NNC: *Packs-sw to high (ventilation). Anti-ice on (improves ENG stall margins). APU on (backup pwr&air).*

29

1

NNC: LAVATORY SMOKE

Smoke is detected in one or more lavatories

@analyse/recap: CC: verify cabin smoke is contained (fire out?)
@airborne: ico smoke-> do NNC Smoke, fire or fumes
Also start diversion until sure fire is contained (mayday)

-Can be a detector fault (if no visible smoke/fire is present but aural wrng continues)
- *DDP Associated lavatory is kept locked exept for use opr inspection of crew members. fire extinguisher bottle position should be aware with cabin crew members.*

29

2

YOU ARE IN NNC:
SMOKE OR FUMES REMOVAL

AND

NOW -->
THE SMOKE OR FUMES BECOMES
THE GREATEST THREAT

NNC: SMOKE OR FUMES REMOVAL

hier kom je alleen via de NNC: smoke, fire or fumes

If needed: Oxygen masks (pilots only!)

STP: should already be diverting

NNC: if CAB ALT WRNG-> disregard, its due to the actions

If no time (imm lnd): 2xUtils off, 2xEquip ALTN, 2xRecirc off, when <3000ft -> even Packs off

Worst case @ open water -> Ditching is an option

29

3

uncommanded
stab trim
movement occur

or in a manner
not appropriate
for flight

NNC:RUNAWAY STABILIZER

LONG FINAL - NO AUTOLAND - NO RVSM

@ T/O <V1 = reject, unsafe to fly
@ climbout: max FL290, "negative RVSM"

Fly the a/c: IAS towards trim-speed is easier man trim
GA review:
- adv thrust slowly to avoid excessive pitchup movmnt
- Req straight ahead climbout
- Long final & configure early

65ton, max REV thr, max manual =
(DRY) 1250m / (GD) 1700m / (GD/MD) 2000m / (MD) 2300m

29

4

Loss of electric trim through the control wheel switches

NNC: STABILIZER TRIM INOPERATIVE

FL15 LND - LONG FINAL - NO AUTOLAND - NO RVSM

@appr: if not critical -> G/A due to possible lack of pitch authority in case of Flare or G/A

GA review:
- adv thrust slowly to avoid excessive pitchup movmnt
- Req straight ahead climbout
- Long final & configure early

65ton, max REV thr, max manual =
(DRY) 1250m / (GD) 1700m / (GD/MD) 2000m / (MD) 2300m

29

5

There is no
(or abnormal)
response
to thrust lever
movement

NNC: ENGINE LIMIT OR SURGE OR STALL

N-1 -> FL15 LAND

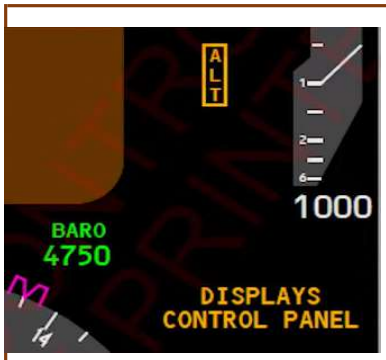
@clb: Level off when appr. FL200
@crz: if needed -> ENG driftdown procedure
@appr: NNC: Turn off engine and prepare N-1 approach
@short final: landing with N-1 or GA N-1

NNC: Check engine driftdown procedure
- NNC engine inop landing checklist
- Check perform. limitat.

Fly the a/c: if yaw moment is uncontrollable due to stuck setting
> use rudder to control yaw, or shutdown engine

30

1



ALTIMETER
BLANKS

+

ALT
FLAG

NNC:DISPLAYS CONTROL PANEL

EFIS control panel is failed (altimeter & minimums are lost)

NO RNP AR - NO AUTOLAND

@gnd= nogo
@fly: consider switching controls to good side *(RVSM=ok)*
@approach: continu but recheck baro-setting
(no x-check possible)
@final: if PF losing ALT indic: switch controls or go-around

NNC: control panel sel sw: BOTH ON 1 or 2 (good side)

- Ensure correct baro-settings when starting approach
- RA keeps working

30

2

+

FMC DISAGREE (message on CDU)

NNC: FMC DISAGREE

Essential info between L/R FMC do not match

@gnd: nogo, due to no crosschecking (essential) possible
@airborne: verify position with raw data (IRS pos)
(FMC navigation check Suppl.Proc)
@RNP appr: Go around, unless conventional b/u (instant&safely)
or visual reference

- Reason to discontinue RNP approach (even after IAF) due to RNP operative eqpmnt fail. Unless visual reference.
- Both FMC remain avlb, however revert to single FMC ops on approach

30

3

FMC DISAGREE - VERTICAL
(message on CDU)

LEFT & RIGHT FMC target airspeeds disagree during descent
-OR-
LEFT & RIGHT FMC vertical paths
disagree during descent.

NNC: FMC DISAGREE - VERTICAL

IAS or vertical path disagreement between FMC's

Monitor ALT restrictions carefully

Pre-stage of FMC DISAGREE
- *problem most likely corrects itself on next WP with an ALT constraint*
- *If remains when starting appr -> it changes to FMC DISAGREE*
- *Do not use FMS sel sw:*
this interrupts possible automatic correction.

30

4

message
SINGLE FMC
OPERATION

NNC: FMC FAIL

NO RNP

One or both FMC's defect (this image = single fail)

1 failed FMC: FMS source select sw to working FMC

ico 2 failed FMC's: *loss of LNAV/VNAV and N1 limit*
@fly a/c: - HDG SEL (LVL CHG) "unable RNAV req vctrs"
- Fly conventional nav. (be TERR aware!)
@clb: A/T has no limit > (consider N1 set slctr / man.thrust)
- NNC: SPD REF sel.: use PI table: 'VREF' (white bug set Vref+20)
N1 SET sel. use PI table: 'Go Around %N1' *or use dynamic source!*
- RVSM and CAT II/III still available (only conventional approach)

30

5

IN FLIGHT:
STEADY WARNING HORN SOUNDS

NNC: LANDING CONFIGURATION

Flaps/Gear not in landing configuration

@appr/ final: Fly the a/c, correct configuration
If not stabilized -> go-around

- Steady horn: FL 15-40 with gear not down&locked or
< 800' RA with low thrust setting or

31

1

CLACKER SOUND

NNC:OVERSPEED

airspeed is more then Vmo/Mmo

Reduce thrust (not to idle, due to slow spool-up) and partial speedbrake deployment.

Note the max. speed (and time) in flightlog.

31

2

Airborne:
Flames in engine inlet or exhaust
(no further fire indication)

NNC: ENGINE LIMIT OR SURGE OR STALL

@clb: Level off when appr. FL200 | if N-1 -> FL15 LAND
@crz: if needed -> ENG driftdown procedure
@appr: if eng indications stabilize after MEM ITEMS ->
consider continue the appr. If needed > go-around N-1
@short final: landing with N-1 or GA N-1

NNC: if continue with reduce thrust: (check eng driftdn procdr)
- consider eng partially inop. - Check perform. limitat.
- consider NNC eng failure as guideline

- if limit exceedence icw vibrations -> NNC ENG SEV DMG
- *Considerations Eng restart attempt: ? only rear cases (Sub's)*
- *EGT 800˚C is abnormal for CRZ, MCT is appr 750 ˚C.*

31

3

NNC: ENGINE LIMIT OR SURGE OR STALL

@clb: Level off when appr. FL200 | if N-1 -> FL15 LAND
@crz: if needed -> ENG driftdown procedure
@appr: if eng indications stabilize after MEM ITEMS ->
consider continue the appr. If needed > go-around N-1
@short final: landing with N-1 or GA N-1

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- consider eng partially inop. - Check perform. limitat.
- consider NNC eng failure as guideline

- if limit exceedence icw vibrations -> NNC ENG SEV DMG
- *Considerations Eng restart attempt: ? only rear cases (Sub's)*
- *EGT 800˚C is abnormal for CRZ, MCT is appr 750 ˚C.*

31

4

2X ND DISP

NNC: FMC FAIL

NO RNP

One or both FMC's defect (this image = dual fail)

ico 1 failed FMC: FMS source select sw to working FMC

ico 2 failed FMC's: *loss of LNAV/VNAV and N1 limit*
@fly a/c: - HDG SEL (LVL CHG) "unable RNAV req vctrs"
- Fly conventional nav. (be TERR aware!)
@clb: A/T has no limit > (consider N1 set slctr / man.thrust)

- NNC: SPD REF sel.: use PI table: 'VREF' (white bug set Vref+20)
N1 SET sel. use PI table: 'Go Around %N1'
or use dynamic source!
- RVSM and CAT II/III still available (only conventional apprch)

31

5

GEAR LEVER
JAMMED IN
OFF POSITION

NNC: MANUAL GEAR EXTENSION

Cond: - ANY LDG GEAR is not down&locked when lever is down
- LDG GR lever is jammed in OFF position

@ APPR and GR LVR JAMMED OFF: abort -> NNC

If lvr is in OFF position --> No nosewheel steering available

Fuel penalty (+50%). ALTernate fuel is affected. If in bad wx conditions, consider divertion prior GR extention to good weather conditions.

Close the door after use!

32

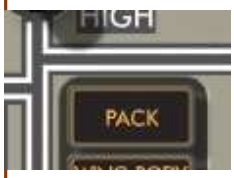
1

hydr **NNC: ENGINE FIRE OR ENGINE SEVERE DAMAGE OR SEPARATION**

FLY A/C: ANC -> RUDDER (lower aileron side)
--> Focus on IAS, remaining eng thrust, yaw,
automation (both A/P) > Then Communicate (PAN/MAYDAY)
@gnd: --> ENG FIRE ON THE GND (backcover)
@TO: --> 12-13gr pitch up, Perform EOP, (MCT), MEM items
@CRZ: -> analyse -> FLY ENG DRIFTD PROC (ANC), MEM items
@final: if stabil-> you can continue. If FL15 -> Vref = FL40+20
Go around: If FL30/40 -> FL15, if FL 15 -> FL 1

32

2



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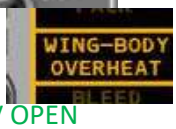


1 PACK LIGHT +
opposite WING ANTI-ICE VLV OPEN
or
opposite WING-BODY overheat

NNC: PACK
+ WING ANTI-ICE VALVE OPEN or WING-BODY OVERHEAT

.

Both want PACK switches off, therefor:
1- retart thrust on respective engine
2- advide ATC immediate descent
3- descent to 10.000 ft or MSA
4- Perform the 2nd NNC
5- Continue unpress. to DEST/ALTN

32

3

+

1 PACK LIGHT +
Opposite
ENGINE FAILURE

.

or
opposite BLEED TRIP OFF

NNC: PACK
+ ENGINE FAILURE OR SHUTDOWN or BLEED TRIP OFF

Both want PACK switches off,
and when ENG FAIL, there is no BLEED
on opposite side due to ISOL VLV CLSD:
1- set ISOLATION VALVE to OPEN
2- Set PACK of ENG FAIL side to AUTO

32

4

Medical situation (cabin)

MEDICAL DIVERSION PANPAN

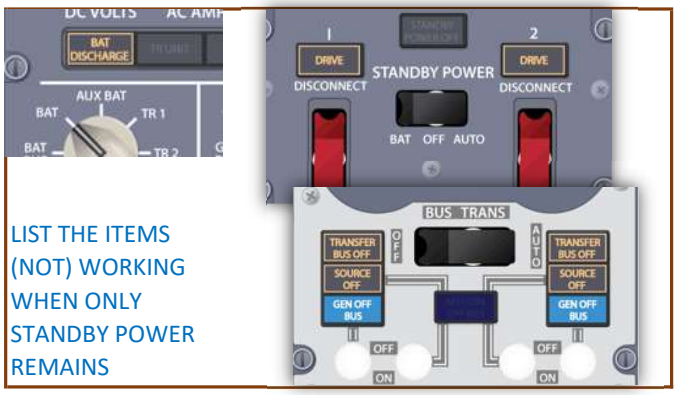
ATC:
-Request Gate which is fast reachable
-Progressive taxi instructions
-Medical crew at the gate
-Ground power source available (bij geen APU).

Brief cabin: Help pax, but be seated when: "take your seats"

PA @ gate: please remain seated, so medical personnel can get on board

32

5



LIST THE ITEMS (NOT) WORKING WHEN ONLY STANDBY POWER REMAINS

NNC: LOSS OF BOTH ENGINE DRIVEN GENERATORS

Items on Standby Power:

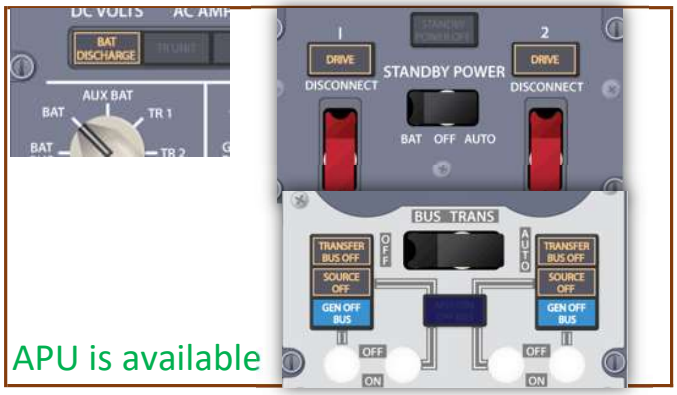
Capt. Pitot probe heat
RH eng ignitor
LH: FMC / CDU / IRS / VHF NAV

No AP / AT (manual trim)	Flaps indic. -> spd tape
60 min battery (APU start -7)	No autobrake
Avoid icing conditions	No Auto speedbrake
Landing flaps 30 (yaw dmpr)	No wipers
Partial Anti-skid	No cab.chime >intercom

Manual pressurization -> open oufl vlv @ pattern alt.

33

1



APU is available

NNC: LOSS OF BOTH ENGINE DRIVEN GENERATORS

check engs for loss of thrust - APU pwr is quickly needed

NO RVSM, NO RNAV, NO AUTOLAND. - LAND PERF

evt by heart GEN switches schakelen + start timer

ENG flameout can occur >30000ft (no powered fuel pumps)

@T/O: no A/T, no elec trim, no FL indic.-> Safe= spd 180kts.

@climb: continue climb max FL250 (apu)

@crz: Gen schakelen, if no luck > MAYDAY + desc. FL250

@final: consider continue VMC & long RWY. -->

No autobrake/antiskid

SUB: Stby pwr bus fail --> avoid icing, use STBY instr. -->

timer clock serves as fuel indicator

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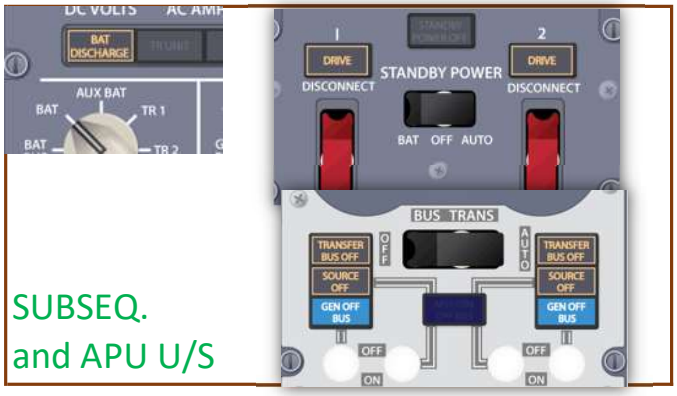
2



SUBSEQ.
and APU U/S

NNC: LOSS OF BOTH ENGINE DRIVEN GENERATORS

check engs for loss of thrust - APU pwr is quickly needed

NO RVSM, NO RNAV, NO AUTOLAND. - LAND PERF

evt by heart GEN switches schakelen + start timer

ENG flameout can occur >30000ft --> MAYDAY + descent

When on stby pwer: NO A/P, NO A/T, NO elec trim,

no probe heat / no (wing) anti-ice / partially anti-skid /

no autobrake/no auto speedbrake/no wipers/no ext.lights

@final: consider continue VMC & long RWY. -->

SUB: APU U/S --> avoid icing conditions, IRS #2 only 5min

check pressurization panel (due to trip off's)

-ALSO 'CABIN ALT WARN OR RAPID DEPRESS' can occur

33

3

2x PACK LIGHT

NNC: PACK
(DUAL PACK LIGHT)

PRESSURIZATION IS LOST

1- Try TRIP RESET

2. NNC

--> Highspeed Descent to MSA or 10.000ft

33

4

Pilot Incapacitation

full
&
partial

Full incapacitation

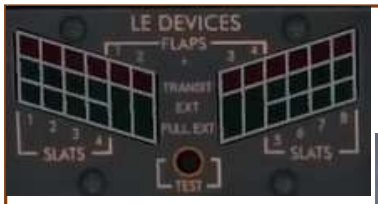
- Take control and command of a/c
- Check essential controls/switches
- Engage autopilot
- "Purser report to cockpit" -Restrain pilot / lock harness
  -slide seat to full aft
  consider help from other pilots/crewmembers aboard
- Mayday call icw medical items (see flashcard)
- Land Nearest Suitable Airport.

Partial incap.

- Be PF, and state who is in control&cmd
- Check essential controls/switches - Engage autopilot
- Notify cabin crew (keep informed)
- Consider notify ATC (evaluate urgency or distress call)
- Consider help from other pilots/crewmembers aboard
- Prolonged flight shall be avoided

33

5



LE FLAPS FAIL
to extend AND
TE FLAPS less than 1

NNC: ALL FLAPS UP LANDING

--> do only when directed by TE Flaps UP LNDG C/L
again only --> when directed by assym/disagree
LND PERF - NO AUTOLAND / CAT 3

Req straight ahead MAP, due to limited bank maneuv.
FL UP man speed = FL 40 + 70kts
Lndg Vref is = FL 40 + 55kts
Tire speed limit 195kts (-1000 KG is -1 kts)
65ton, max REV thr, max manual =
(DRY) 1533m / (GOOD) 1955m / (MEDIUM) 2712m
Not dry/good = Emergency landing
PA: high spd & firm , emerg servics to expect after lndg
Hot brakes 15min after LND, ask for fire brigade to monitor

34
1

REJECTED Takeoff

Before 80kts: Reject for all significant malfunctions
80 kts - before v1: Fire or fire warning, Engine failure,
pred. windsh, or unable/unsafe to fly

Announce REJECT, I HAVE CNTRLS, Disengage A/T,
close THR lvrs, pull speedbr man, apply max rev thrust.

After stop: "CC @ your stations" / "CC norm proc.'

AOR: CPT has control stand.

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2

ENGINE DRIFTDOWN PROCEDURE

Disconnect A/T - Select MCT (FMC) -
manually set limit thrust

Select ENG OUT CRZ:
-> MCP set ALT &IAS -> set LVL CHG -> set CON N1 thrust

ATC -> MEM ITEMS (if appl) -> Turn / short term plan
-> do appliclable NNC

@LVL OFF: MCT - accel SE LRC SPEED - Long term plan

34
3

Fuel considerations

crs: 2400kgs per hour
200 kgs per 5 min

30 min (fin res) is appr 1200 kgs

If holding and 15 min approach = 600 kgs.
600 + fin res 1200 = 1800 kgs.
I want to start approach
latest @ 1800 kgs = xxx minutes
exampl (fuel = 3.0), 1200 kgs extra = 30min bef start appr

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4

Presence of vapour / other hazards during refueling
with Pax on board.

Fasten belts sign should be -OFF
There should already be 2way comms with supervisor of
refueling proces

Fueling should be stopped immediately

Crew: prepare for imm evac -> Consider arm slides of all
closed exits which are not connected to stairs/jetway

34
5

Cold wx ops

Definition and important items

DEF: When operating with low temps, AND with
ice, snow, slush, or standing water present on ramps/twy/rwy
Snowtam
slippery rwy -> reversers important
OAT 3˚ or lwr: every 30 min. runup to 70% (if able) N1 for 30 sec.

Preflight: Probe heat - on
Eng start: ENG AA - on, Wing AA - (on, if no type 2/4 fluid)
Bef taxi: delay FLT cntrl check till after de-icing,
Flaps full range-> then UP till RWY
Taxi: Slow, diff thrust
Bef T/O: Flaps set. ENG runup 40-70% N1, check vibr, display, GO)
AFT lndg: FL 15 till checked@ramp

35
1

v1 / v2

v1 = takeoff decision speed

v2 = minimum speed after liftoff, to safely climb N-1
(meets climb gradient), and
reaching 35 ft at the end of the required runway distance.

Is also the best N-1 angle of climb speed, and is a
minimum spd for flight in that condition until at least
400 ft above the ground.

35
2

PA unruly

PA EMERGENCY LANDING

L&G, here is your cpt speaking. Everybody sit down now and comply to the cabin crews instructions! Anybody who does not comply wil be restrained and arrested upon landing.

L&G here is your CPT speaking, may i please have your attention. We have a technical malfunction with the LDG GR. Because of this, we will perform an emergency landing. The situation is under control, we are fully prepared, and the emergency services on the ground are standing by. Ill ask your attention to the cabin crews instructions to prepare for this emergency landing. Thank you for your attention.

35
3

When immediate landing?

- Bomb (code red or when an object is actually found)
- Fire
- Smoke, fire of fumes become uncontrollable
- Minimum fuel
- Other cases of immediate danger like
  - danger of losing a/c control or
  - imminent airplane brakeup)

35
4

Abundant fuel spilling (on ground)

Stop fueling process

If boarding -> command boarding-staff to stop boarding

If pax on board -> warn persons on board -->
they disembark via norm exits, if no stairs/jetway ->
slides are next normal exits

Shutdown EXT PWR unit, and APU (using norm s/d proc.)

Warn fire brigade

35
5

WHEN ENGINE TAI ON?

WHAT IS CONSIDERED ICING CONDITIONS?

WHEN ICING CONDITIONS EXISTS OR ARE ANTICIPATED, EXCEPT DURING CLIMB AND CRUISE BELOW -40 SAT

ICING CONDITIONS: EXISTS WHEN OAT (GND) OR TAT(FLIGHT) IS 10C OR BELOW AND ANY OF THE FOLLOWING IS PRESENT:

- VISIBLE MOISTURE (*CLDS, FOG WITH VIS 1600m OR LESS, RAIN, SNOW, SLEET, ICE CRYSTALS, AND SO ON...*) OR...
- ICE, SNOW, SLUCH OR STANDING WATER IS PRESENT ON THE RAMPS, TAXIWAYS, OR RUNWAYS

CSFF (cold soaked fuel frost). What is allowed?

- upper wing surfaces?
- lower wing surfaces?

(what is the exception on lower surfaces?)

Upper wing: ALL conditions must be met!

- CSFF is only within the CSFF box
- OAT = 4 degr or higher
- No visible moisture is present
- Tank fuel temperature = -16 degr or higher

(if frost is outside meas.stick 4 -> indicator frost on upper surfaces)

Lower wing: ALL conditions must be met!

- CSFF is only within the CSFF box
- OAT = 4 degr or higher
- No visible moisture is present
- Tank fuel temperature = -16 degr or higher
- Free of snow/ice/frost: LE devices,cntrl surfaces,tab surfaces,winglets

(if Lower wing conditions are not met, light coatings upto 3mm on lower is allowed, unless its on the earlier mentioned surfaces

Tire failure

FCTM has lndng techniques "tire failure" & "lndg on flat tire"

@Takeoff: ONLY RTO <80kts, stopping distance may be effected
retract normally. If lever does not move up, do that NNC.
Inform TWR of poss blown tire.
Consider proceeding destination unless indications of damage/vibr/leaks etc

- Consider long and wide RWY

@Landing: technique normal/smooth
@taxi: normal if no vibration/failures

Emergency landing variants?

How (CRM procedure with cabin crew)

and

Situations in which you prepare the CC for an "emergency landing"

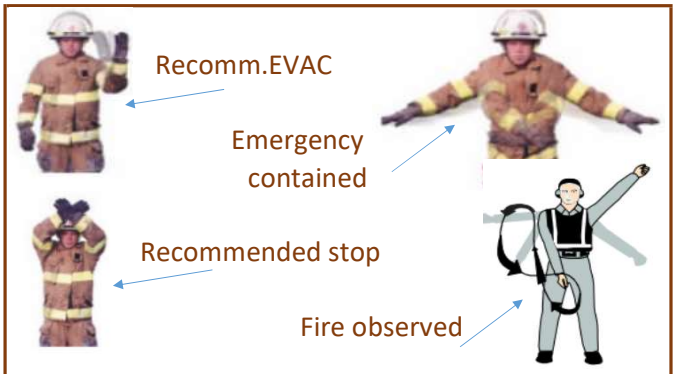
Variants: - Unprepared *(No preparation time)*
- Short prep time <15min
- Long prep time 15-30min

"*Purser report to cockpit* " *(if inop: 3x cabin call button)* -->
"*Prepare for emerg. lndg* "/"*Prepare for ditching* " + WWW
@10000': CC, prepare for landing - @2500' CC, Take your seats
@500' Brace for impact, Brace for impact

Any situation unsure of safe landing,
or when unsure to deboard with stairs/gate.
Ditching / Bomb incident / Glide in / Fire / Tire burst / Jammed or restricted flight controls / Fuel low condition / Manual reversion / Partial or all gear up landing / Tank leak.

Fire brigade hand signals

- Fire observed
- Recommended evacuation
- Recommend stop
- Emergency contained



IF RNP operative equipment fail
'AFTER' the
IAF
during RNP approach

what to do?

Approach may be continued provided:

- Visual reference is established or
- Navigation performance is sufficient:
  ANP is within RNP
  - i.e. 0.3 in final segment,
  - 1.0 in initial/intermediate/MAP segment

Discontinue if:
- UNABLE REQD NAV PERF-RNP
- FMC DISAGREE
- any message: VERIFY POS XXX_XXX

37
1

Extra considerations when….
Loss of SYSTEM B

and now this comes up..

STANDBY HYDRAULIC LOW PRESSURE

Only 1 hydr system left > **Land at nearest suitable airport**

@ final and time/fuel critical:
-continue if long RWY, add 15kts to FAS

- Loss of thrust reverser #2,
- LE devices will not extend at all. -->
  "LEADING EDGE FLAPS TRANSIT" light is result.
  Light comes on after setting TE FL 15.
  Expect 15 kts higher FAS. (FL 15 + 15kts)

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2

Extra considerations when….
Loss of SYSTEM A

and now this comes up..
STANDBY HYDRAULIC LOW PRESSURE
or
dispatched with ALT NOSEWHEEL STEERING INOP

Only 1 hydr system left > **Land at nearest suitable airport**

STBY HYD LOW PRESSURE:
- Loss of thrust reverser #1,

Dispatched with ALT NOSEWHEEL STEERING INOP:
- NO nosewheel steering!
  Avoid wet RWY with crosswind
  Below 60kts use differential braking
  Towing assistance after landing

37
3

Transfer bus 1 failed items

NO ILS / NO AUTOLAND / avoid ICING
isol valve inop
Multiple pitot heat: avoid icing conditions
Left wing anti-ice vlv stuck in last position
Window heat all LH windows consider FO PF
A/P A
A/T retard mode
ATC 1 inop no TCAS
SYST B electr hyd pump
GPWS
APP mode - DC bus 1 (cross tie relay) Bars retract @GS capture
Raw data ILS can be perf LNAV/VNAV with ILS tuned

37
4

Transfer bus 2 failed items

NO ILS / NO AUTOLAND / avoid ICING / FL15 LND
duct press indication
Multiple pitot heat: avoid icing conditions
A/P B (A/P A after 5min, loss of IRS#2)
TE flap indicator Use speed tape indications
ALT flap extension Use normal flaps
Electric trim A/P trim is avail. Use A/P if poss.
STAB TRIM INOP --> DO this NNC --> FL 15 lndg
Rudder trim indicator
Nav control panel #2
SYST A electric hyd pump
Standby hyd pump

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5

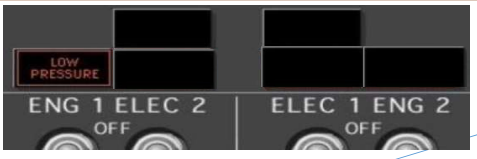
STABILIZED CRITERIA FOR
VISUAL APPROACHES

STABILIZED CRITERIA
AT THRESHOLD

at 100' AFE, the a/c should be positioned within, and tracking within, the lateral confines of the twy edges extended.

Stabilized not lower then 1000' AFE during IMC / intcpt from above proc.
Stabilized not lower then 500' AFE during VMC
VMC: (5km / no clds below 1000') - 3.7km for circling appr.

- tracking req lateral & vertical path
within 1 dot GS/GP (or full exp loc scale) - or 3reds-whites
- at Appr/tgt speed (+10 / -5 kts) and trending good direction
- RoD is 1000fpm or less, unless briefed - AOB less then 30degr
- correct configuration and c/l performed @500ft
- thrust setting is appropriate/stabilized
- a/c in correct configuration
- @circling: wings level @ 300'AFE
- POSITIONED TO MAKE A NORMAL LANDING IN THE TDZ
(3000FT OR 1ST THIRD OF RWY >LOWEST)



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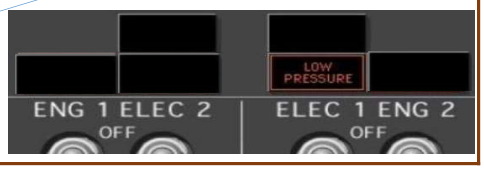

NNC: HYDRAULIC PUMP LOW PRESSURE
can be the engine-driven-pump EDP or
electric-motor-driven pump EMDP

@analyse: check hydr page for leaks
can be combination with gear disagree
in that case -> perform this NNC 1st

NNC: set switch to OFF
during Goaround: flaps&gear can move @ slower rate

- ddp: nogo
- if 1 remaining EDMP: may flicker during high demands